

Table 1.3. Result of Trend Analysis for the considered gauging stations in UKB.

Gague name	Gauge ID	Area	Years	PETTITT p-value	PETTITT trend	PETTITT year	CUSUM p-value	CUSUM trend	CUSUM year	MK p-value	MK trend	Is Shift Auto	Is Shift Manual	Year Match
Karad	10001	1550	59	0.108591	NEGATIVE	1997	0.024547	NEGATIVE	2021	0.030925	NEGATIVE	0	0	0
Warunji	10002	389	58	0.524983	POSITIVE	2003	0.206897	NEGATIVE	2021	0.830036	NEGATIVE	0	0	0
Samdoli (Seasonal)	10004	2850	57	0.010075	NEGATIVE	1994	0.004592	NEGATIVE	1994	0.030127	NEGATIVE	1	0	1
Arjunwad (Seasonal)	10005	3760	55	0.078556	NEGATIVE	1997	0.110154	NEGATIVE	2008	0.039235	NEGATIVE	0	0	0
Terwad (Seasonal)	10006	2560	45	0.348078	POSITIVE	2015	0.137069	POSITIVE	2015	0.898806	POSITIVE	0	0	1
Kurundwad	10007	6470	52	0.265403	NEGATIVE	1997	0.5	NEGATIVE	2022	0.339669	NEGATIVE	0	0	0
Sadalga (Seasonal)	10008	7740	55	0.003086	NEGATIVE	1991	0.00027	NEGATIVE	1994	0.009352	NEGATIVE	0	0	0
Gokak Falls (Seasonal)	10009	187	53	0.001674	NEGATIVE	1997	0.000152	NEGATIVE	2008	3.56E-05	NEGATIVE	0	0	0
Mudhol	10010	5069.9	17	0.263597	NEGATIVE	2019	0.066387	NEGATIVE	2019	0.483756	NEGATIVE	0	0	1
Koyna	10011	5436.4	36	0.570246	POSITIVE	1974	0.055556	NEGATIVE	2006	0.692839	POSITIVE	0	0	0
Sangali (Irwin Bridge)	10012	-	19	0.340613	NEGATIVE	2008	0.421053	NEGATIVE	2021	0.36302	NEGATIVE	0	0	0
Rajaram KT weir	10013	-	36	0.208135	POSITIVE	2015	0.104241	POSITIVE	2015	0.653079	POSITIVE	0	0	1

From Table 1.3, it can be inferred that the Pettitt p-value calculated from the test statistic is less than a chosen significance level (e.g., 0.05) in case of gauging stations Samdoli, Sadalga and Gokak Falls, that the null hypothesis is rejected for these gauges, indicating evidence of a significant change point. In case of remaining gauges, Pettitt p-value calculated from the test statistic is greater than a chosen significance level, that mean the null hypothesis is accepted, indicating no evidence of a significant change point. The Pettitt trend test results indicates that the Warunji, Terwad, Koyna and Rajaram KT weir gauging sites having positive trend whereas other remaining gauging sites having the negative trends. The Pettitt year column in Table 1.3 shows the change point year for that particular gauging site. CUSUM-trend test shows the positive trends on for gauging sites Terwad and Rajaram KT weir. The Mann-Kendall p-values calculated from the test statistic is less than a chosen significance level (e.g., 0.05) in case of gauging sites Karad, Samdoli, Arjunwad, Sadalga and Gokak Falls, that mean the null hypothesis is rejected for these gauges, indicating evidence of a significant trends in time series data. The Mann-Kendall trend test value presented in Table 1.3 suggests that the trend for the gauging sites Terwad, Koyna and Rajaram KT weir is positive (mean increasing) otherwise it is negative for the remaining gauging stations. The results of trend analysis presented in Table 1.3 for all the considered gauging sites can be visualized graphically from Figures 1.4 to 1.15.

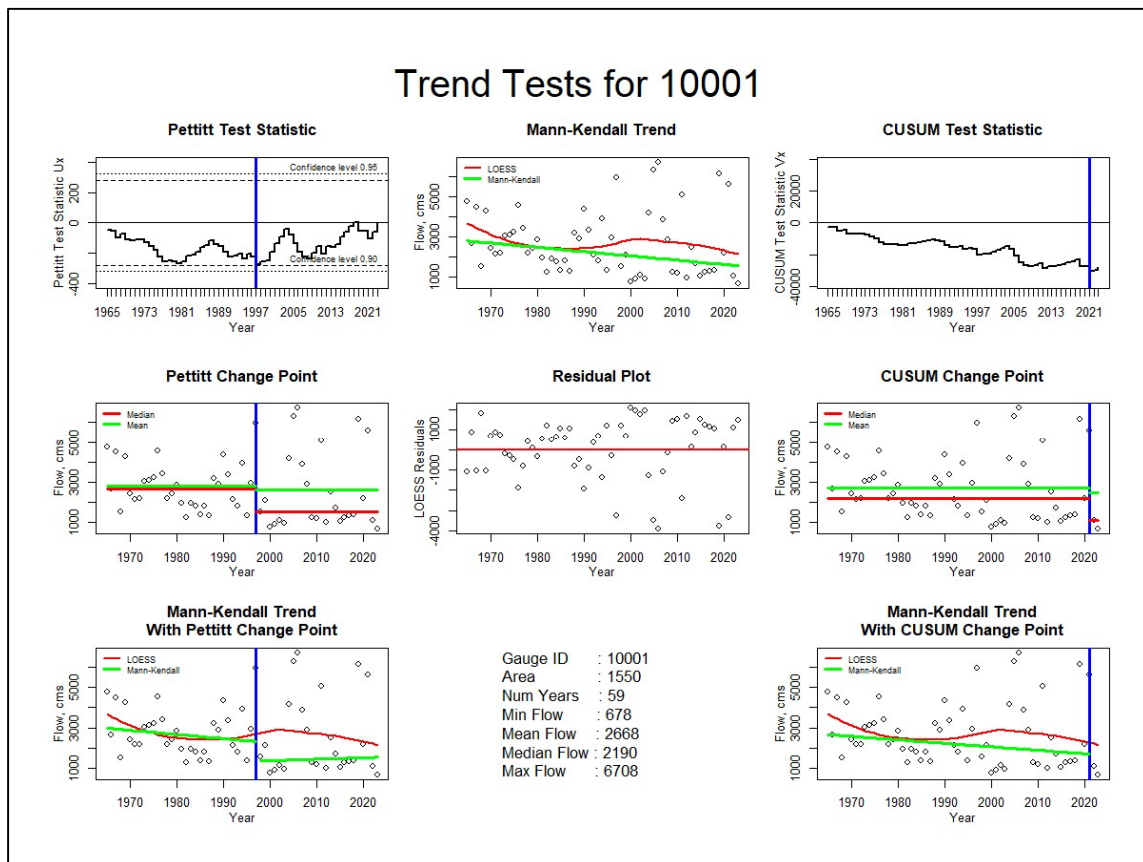


Figure 1.4. Trend tests results for Karad Station (Gauge ID: 10001)

Trend Tests for 10002

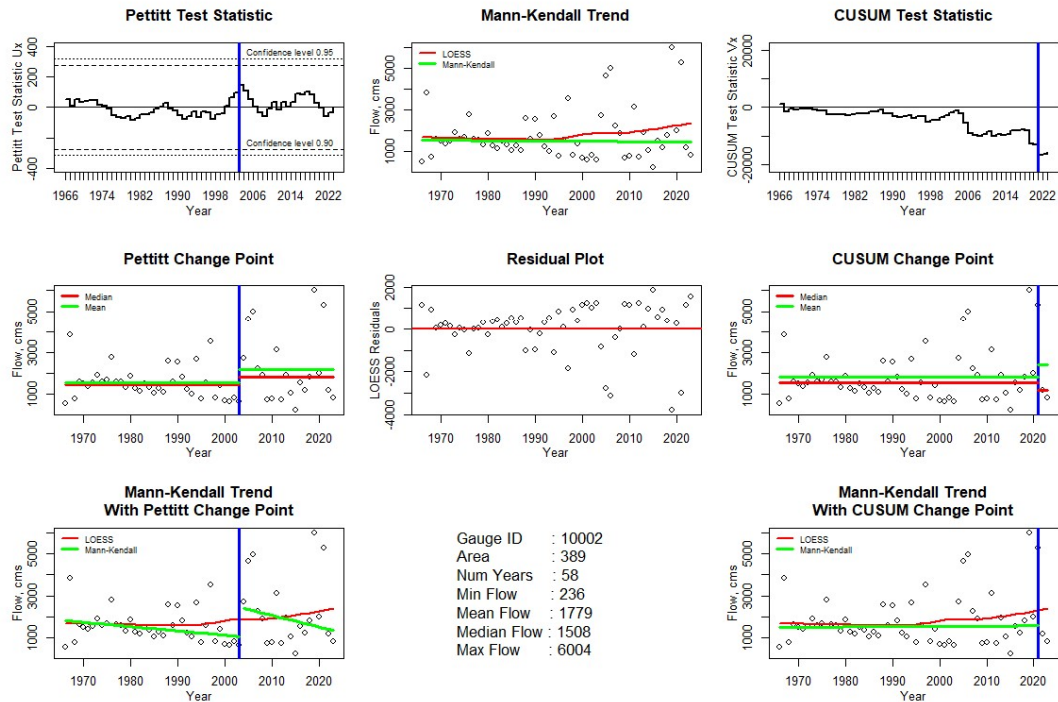


Figure 1.5 Trend tests results for Warunji Station (Gauge ID: 10002).

Trend Tests for 10004

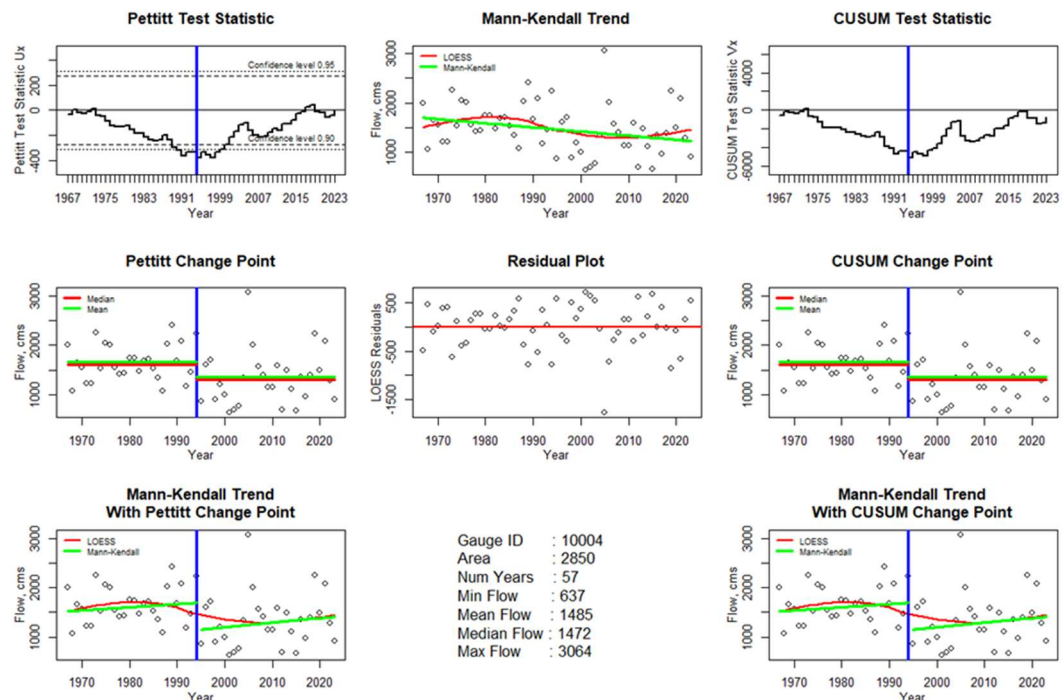


Figure 1.6 Trend tests results for Samdoli Station (Gauge ID: 10004)

Trend Tests for 10005

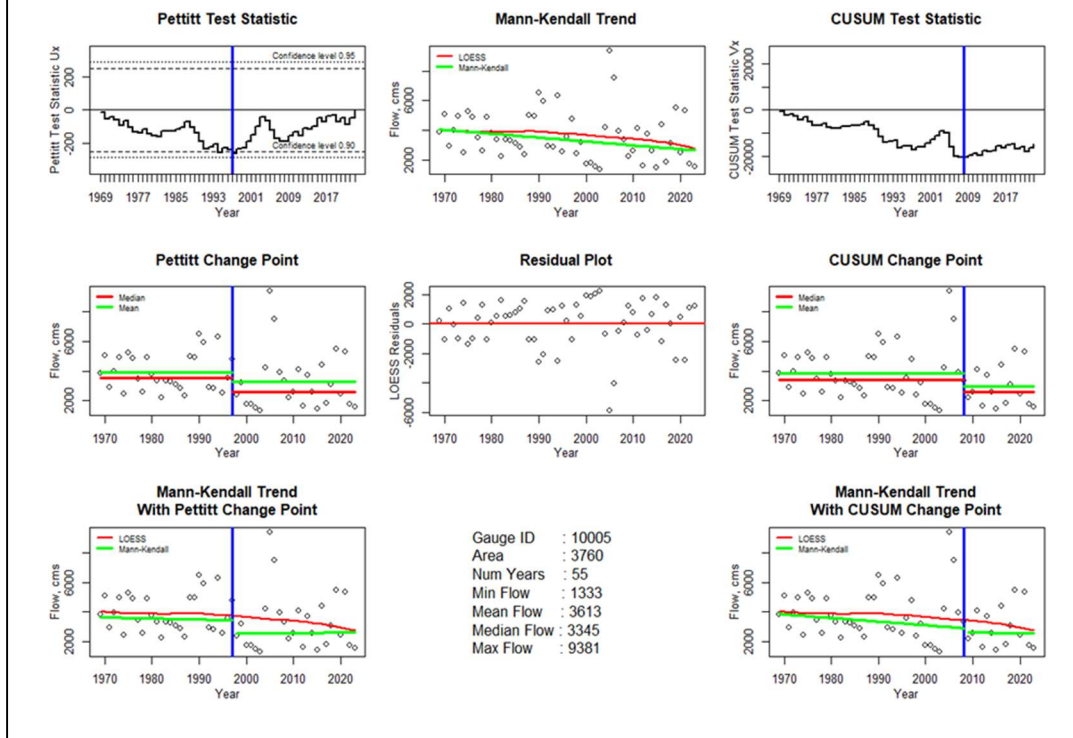


Figure 1.7 Trend tests results for Arjunwad Station (Gauge ID: 10005)

Trend Tests for 10006

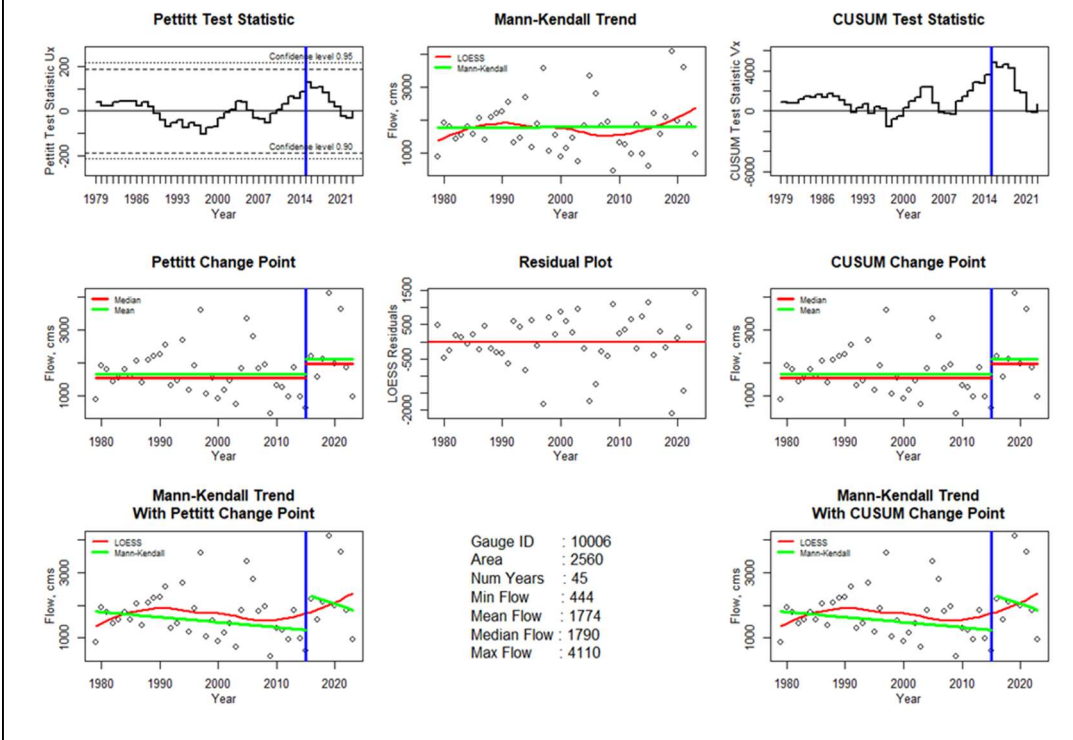


Figure 1.8 Trend tests results for Terwad Station (Gauge ID: 10006)

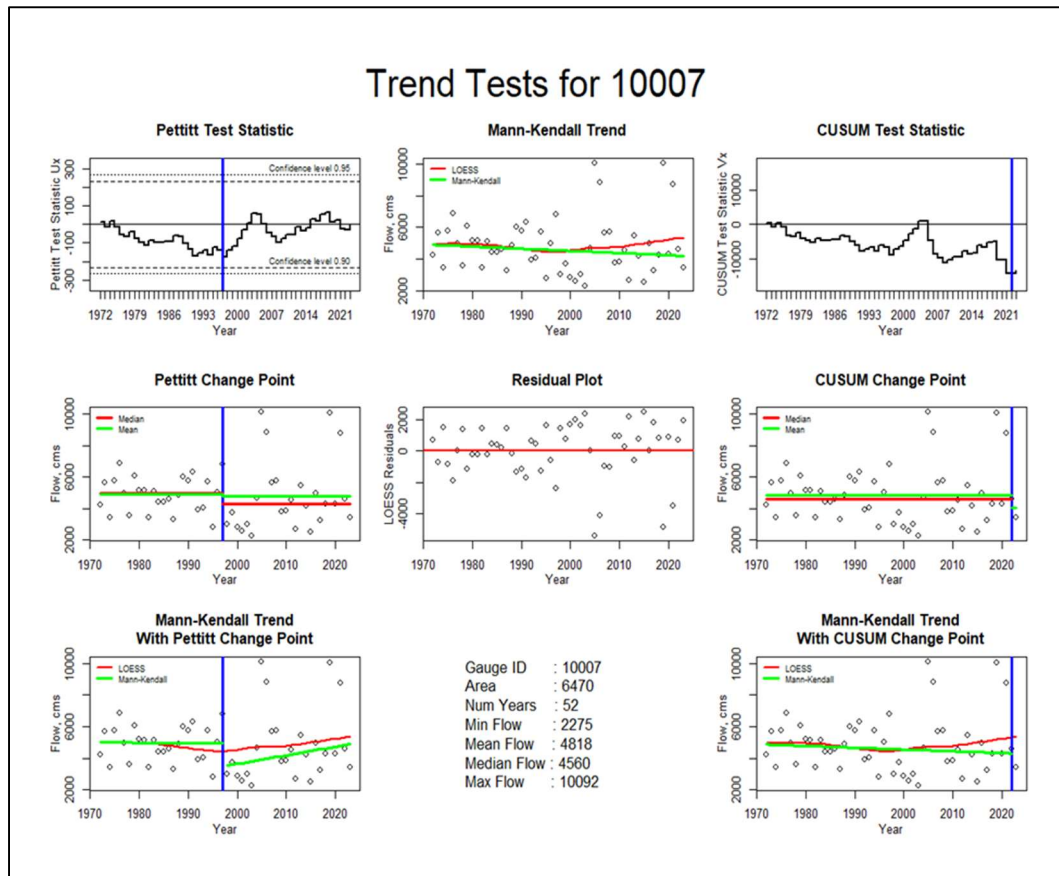


Figure 1.9 Trend tests results for Kurundwad Station (Gauge ID: 10007)

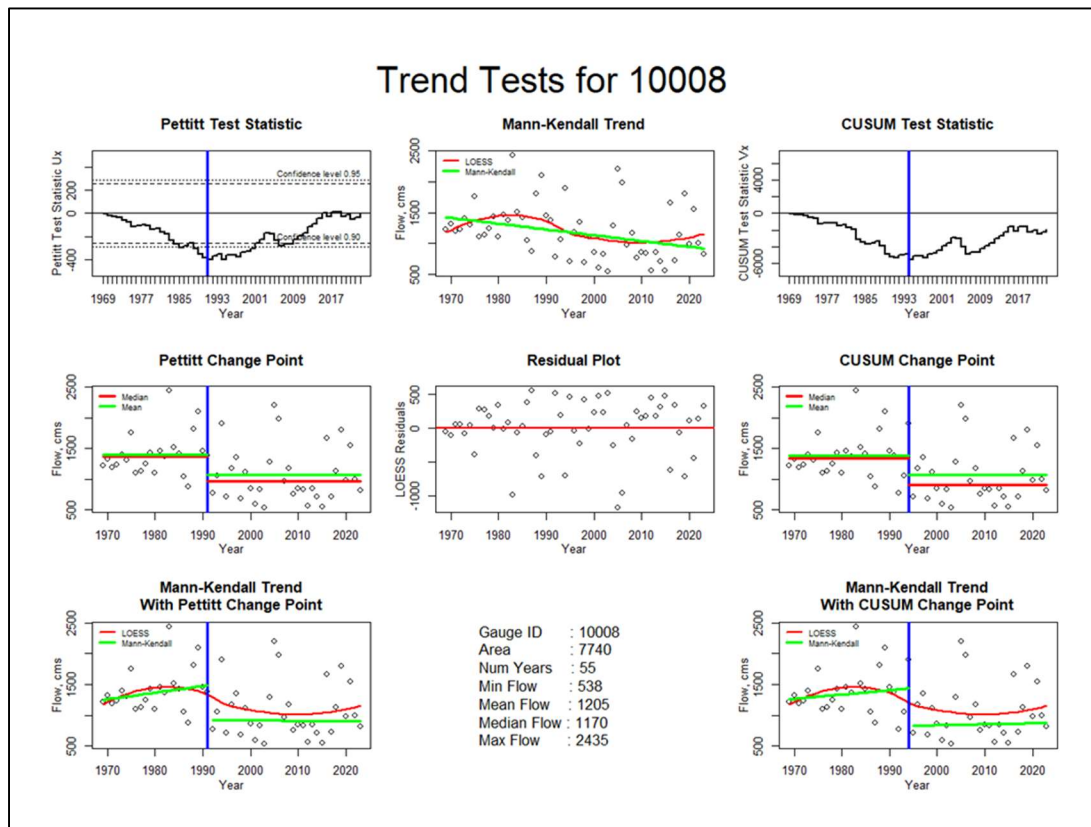


Figure 1.10 Trend tests results for Sadalga Station (Gauge ID: 10008)

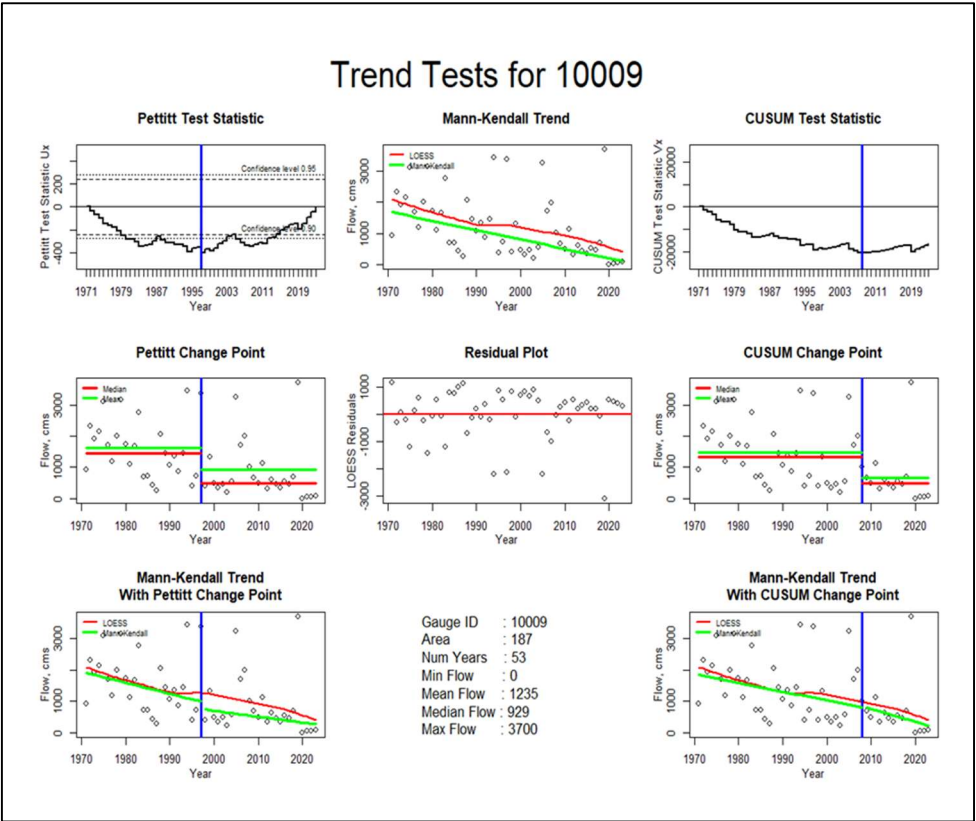


Figure 1.11 Trend tests results for Gokak Falls Station (Gauge ID: 10009)

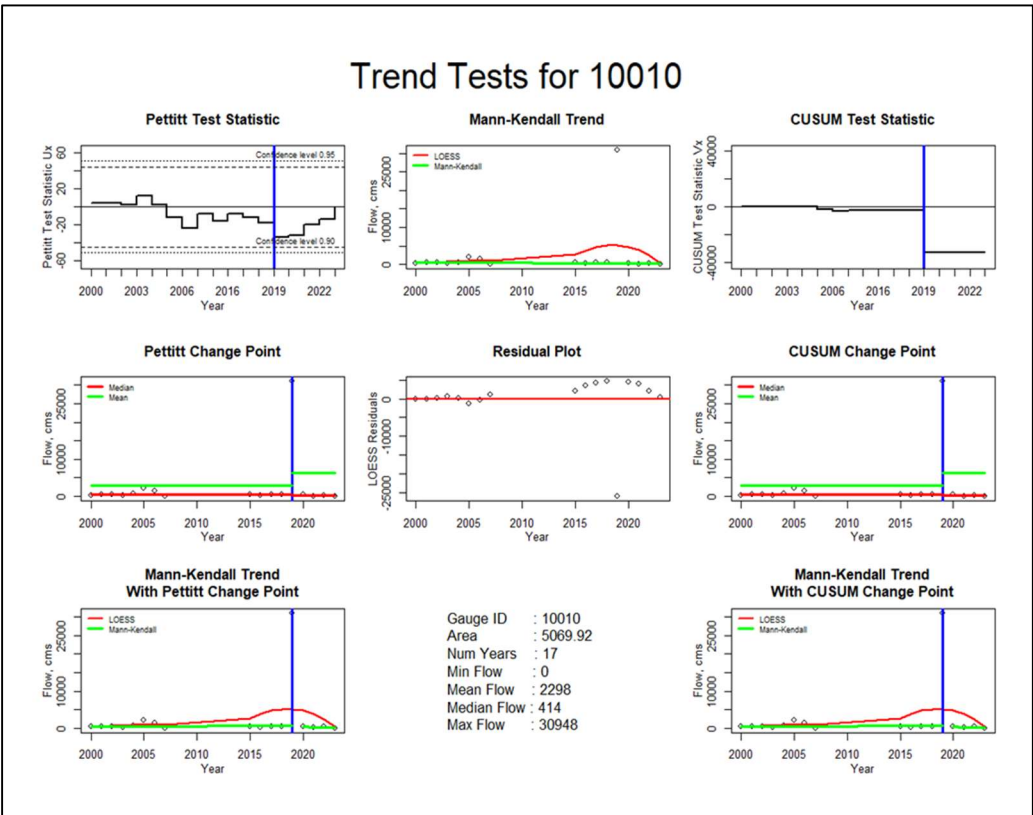


Figure 1.12 Trend tests results for Mudhol Station (Gauge ID: 10010)

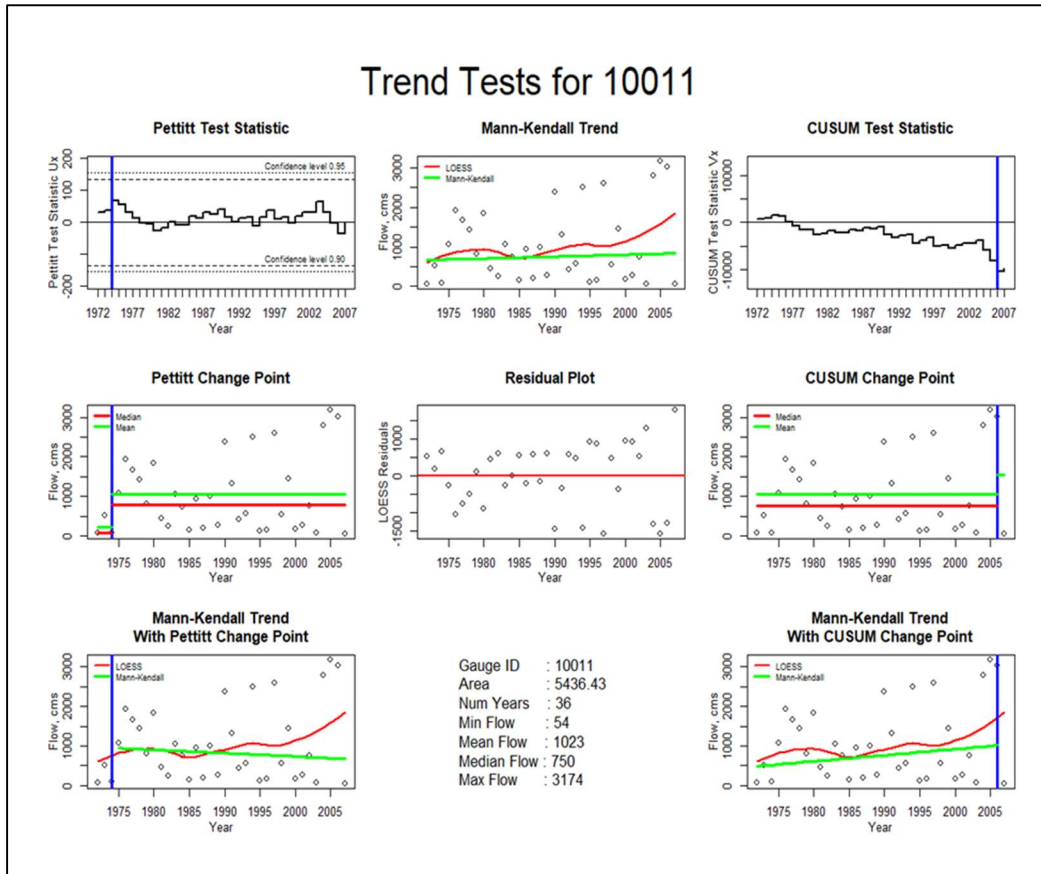


Figure 1.13 Trend tests results for Koyna Station (Gauge ID: 10011)

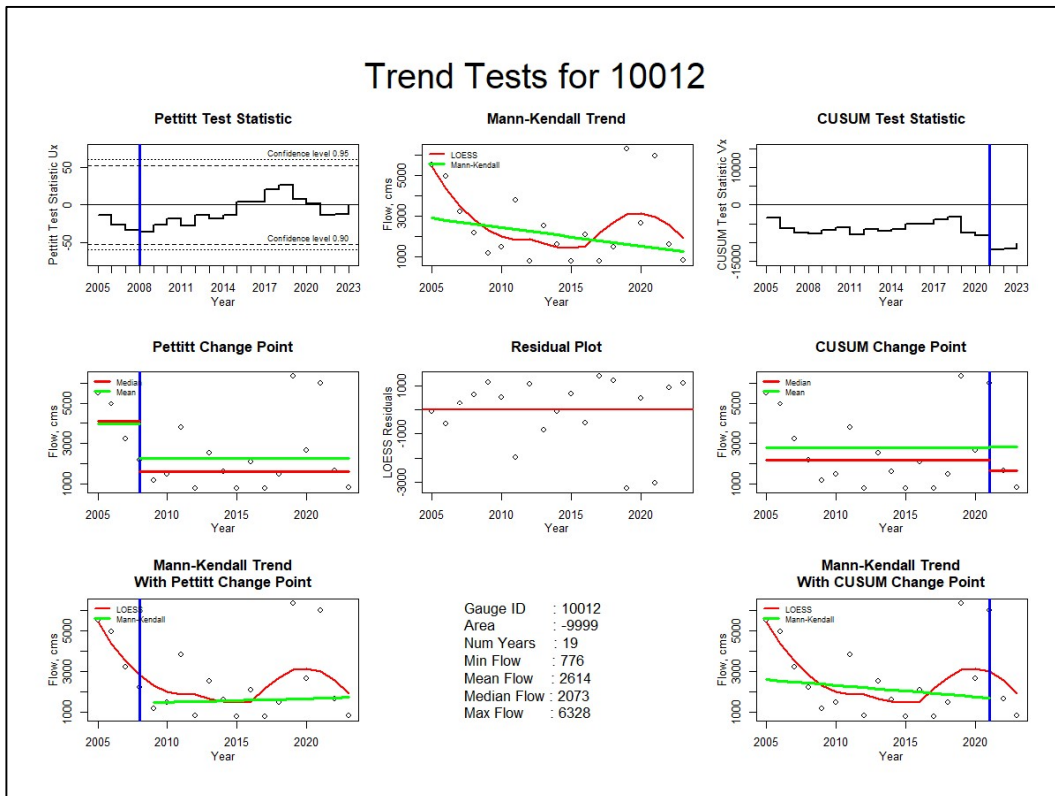


Figure 1.14 Trend tests results for Sangali (Irwin Bridge) Station (Gauge ID: 10012)

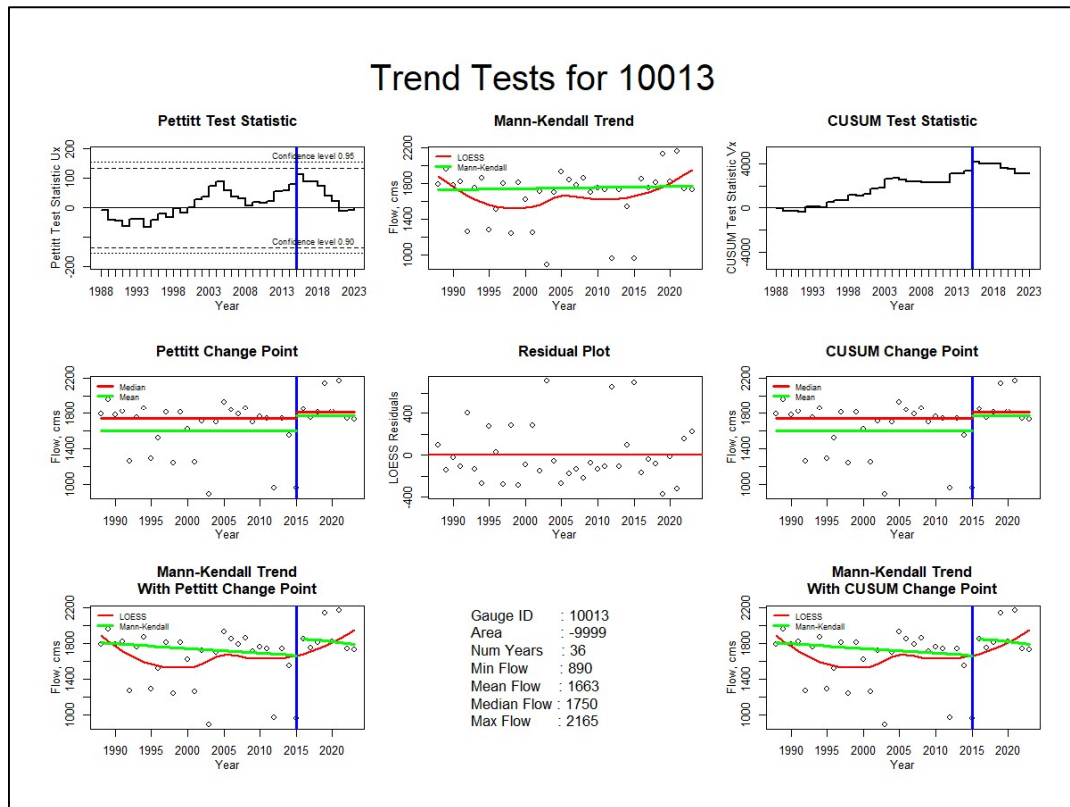


Figure 1.15 Trend tests results for Rajaram KT weir Station (Gauge ID: 10013)

Statistical Properties of the Data

Each of the statistical properties of annual maximum flood data has specific significance and provides valuable insights into the nature and behavior of the dataset. The statistical properties calculated for each gauging sites is presented in Table 1.4. A detailed explanation of the significance of each statistical property can be explained as follows:

Mean of Sample:

- **Significance:** The mean provides the average value of the annual maximum flood data. It is a measure of central tendency, indicating the typical magnitude of the annual maximum floods.
- **Interpretation:** A higher mean suggests that the floods are generally larger, while a lower mean indicates smaller flood magnitudes. From Table 1.4, mean of sample at gauging site Kurundwad is higher as compared to other gauging sites which suggests flood magnitudes is larger. The maximum mean value of 4818.3 Cumec observed at Kurundwad whereas minimum mean value 731.2 Cumec observed at Mudhol.

Variance of Sample:

- **Significance:** Variance measures the spread or dispersion of the data points from the mean. It quantifies how much the annual maximum flood values vary.
- **Interpretation:** A high variance indicates that the flood magnitudes vary widely from year to year, whereas a low variance suggests that the magnitudes are relatively

consistent. From Table 1.4, variance of sample estimated at gauging sites Sangali, Kurundwad, Arjunwad, Karad and Warunji, respectively are higher as compared to other gauging sites which suggests flood magnitudes vary widely at these gauging stations. The variance estimated for Terwad station is higher than Samdoli. A variance of sample estimated at Rajaram KT weir and Sadalga gauging sites are low which means that the flood magnitudes are relatively consistent at these gauging stations.

Standard Deviation:

- **Significance:** Standard deviation is the square root of the variance and also measures the dispersion of the data points. It is expressed in the same units as the data, making it more interpretable.
- **Interpretation:** Similar to variance, a high standard deviation indicates large fluctuations in flood magnitudes, while a low standard deviation indicates more consistency. From Table 1.4, standard deviation of sample estimated at gauging sites Sangali, Kurundwad, Arjunwad, Karad and Warunji, respectively are higher as compared to other gauging sites which suggests flood magnitudes vary widely at these gauging stations. A standard deviation of sample estimated at Rajaram KT weir and Sadalga gauging sites are low which means that the flood magnitudes are relatively consistent at these gauging stations.

Maximum of Values:

- **Significance:** The maximum value represents the highest annual maximum flood observed in the dataset.
- **Interpretation:** It indicates the most extreme flood event within the period of record. From Table 1.4, the maximum discharge values observed at Kurundawad, Arjunwad, Karad and Sangali (Irwin bridge) gauging sites were 10092, 9381, 6708.2 and 6327.8 Cumec, respectively. Whereas the maximum discharge values observed at Terwad, Samdoli and Sadalga were 4110, 3064 and 2435 Cumec, respectively.

Minimum of Values:

- **Significance:** The minimum value represents the lowest annual maximum flood observed in the dataset.
- **Interpretation:** It indicates the least extreme flood event within the period of record. Table 1.4, the minimum discharge values observed at Karad, Sangali (Irwin bridge), Arjunwad and Kurundawad gauging sites were 677.6, 775.5, 1332.6, and 2275 Cumec, respectively. Whereas the minimum discharge values observed at Samdoli, Terwad, and Sadalga were 4.1, 444.3, and 537.7 Cumec, respectively.

Median of Values:

- **Significance:** The median is the middle value when the data is sorted in ascending order. It is a robust measure of central tendency that is not affected by outliers.
- **Interpretation:** The median provides a better measure of the typical flood magnitude when the data contains extreme values or is skewed. The median value of the Kurundwad gauging site is higher as compared to the other gauging sites.

Table 1.4 Statistical Properties of the Station Data

Station/ Property	Mean	Variance	Standard Deviation	Maximum of Values	Minimum of Values	Median of Values	Coefficient of Variation	Coefficient of Skewness	Coefficient of Kurtosis
Karad	2668	2420039.1	1555.6	6708.2	677.6	2190	0.58	0.94	0.08
Warunji	1779.1	1454226.3	1205.9	6003.7	235.7	1507.8	0.68	1.79	3.25
Samdoli	1411.7	331889.1	576.1	3064	4.1	1449.2	0.41	-0.15	0.94
Arjunwad	3612.6	2692486.9	1640.9	9381	1332.6	3345.3	0.45	1.09	1.7
Terwad	1774.3	649156.4	805.7	4110	444.3	1790	0.45	1	1.12
Kurundwad	4818.3	3133510.2	1770.2	10092	2275	4560.4	0.37	1.26	1.88
Sadalga	1204.7	192408	438.6	2435	537.7	1170	0.36	0.71	0.23
Gokak Falls	1234.9	1007351.2	1003.7	3700	0	929.1	0.81	0.96	-0.02
Mudhol	731.2	848421.8	921.1	3585.3	24.7	416.5	1.26	2.44	6.08
Koyna	1023	891875.6	944.4	3174	53.9	749.8	0.92	0.93	-0.28
Sangali (Irwin Bridge)	2614.4	3407494	1845.9	6327.8	775.5	2072.8	0.71	0.95	-0.39
Rajaram KT weir	1663.1	92998.9	305	2165	890	1750.5	0.18	-1.16	0.98

Coefficient of Variation (CV):

- **Significance:** The CV is the ratio of the standard deviation to the mean, expressed as a percentage. It provides a normalized measure of dispersion relative to the mean.
- **Interpretation:** A higher CV indicates greater variability relative to the mean, while a lower CV indicates more consistency. From Table 1.4, the Mudhol gauging site has higher CV that mean it has greater variability relative to the mean. Whereas, the Rajaram KT Weir gauging site has lower CV that mean it has low variability relative to the mean.

Coefficient of Skewness:

- **Significance:** Skewness measures the asymmetry of the data distribution. It indicates whether the data is skewed to the left (negative skew) or to the right (positive skew).
- **Interpretation:** A positive skewness indicates that the data has a longer right tail, with more extreme high values. A negative skewness indicates a longer left tail, with more extreme low values. A skewness close to zero indicates a symmetric distribution. From Table 1.4, the Rajaram KT Weir and Samdoli gauging sites have negative skewness which indicates a longer left tail, with more extreme low values. The Mudhol, Warunji, Kurundwad, Arjunwad, Terwad and Karad gauging sites, respectively shows positive skewness with higher values that mean they have a longer right tail, with more extreme high flood values.

Coefficient of Kurtosis:

- **Significance:** Kurtosis measures the "tailedness" or peakedness of the data distribution. It indicates whether the data has heavy tails (leptokurtic) or light tails (platykurtic) relative to a normal distribution.
- **Interpretation:** A high kurtosis value indicates more frequent extreme values (heavy tails), suggesting a higher risk of extreme flood events. A low kurtosis value indicates fewer extreme values (light tails), suggesting a lower risk of extreme events. From Table 1.4, the Mudhol, Warunji and Kurundwad, Arjunwad and Terwad gauging sites has high kurtosis value indicates more frequent extreme values as compared to other gauging sites.

Distribution Fit Analysis

In this analysis, we compare different statistical distributions to determine which best fits a given dataset. The comparison is based on several goodness-of-fit (GoF) criteria: AIC, AICc, BIC, ADC, KS, AD, and a Comb Criterion. The results of distribution fit analysis for all considered 13 gauging station using various GoF criteria, when the Maximum Likelihood Method and Method of Moments are presented in Figures 1.16 and 1.17, respectively. From these figures, it can be inferred that the P3, LN and LP3 are the most suitable probability

distribution for almost all the gauging sites based on used various GoF criteria. Further, as seen from Figures 1.18 and 1.19, the GEV and GUMBEL were highest GoF passing rate for when GoF criteria such as KS, AD and combination of these two were used.

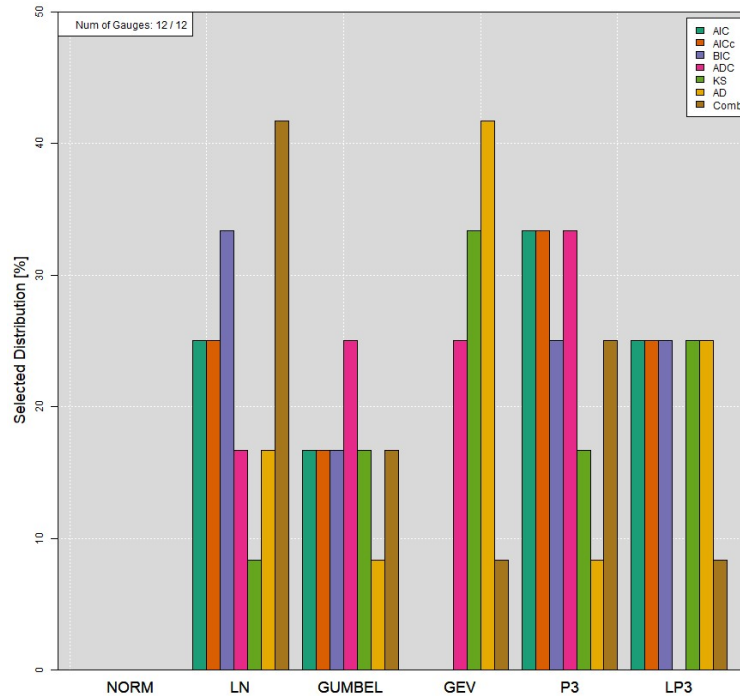


Figure 1.16 Distribution fit analysis for all considered 13 gauging station using various GoF criteria when the Maximum Likelihood Method was used to estimate the parameters of the various distributions.

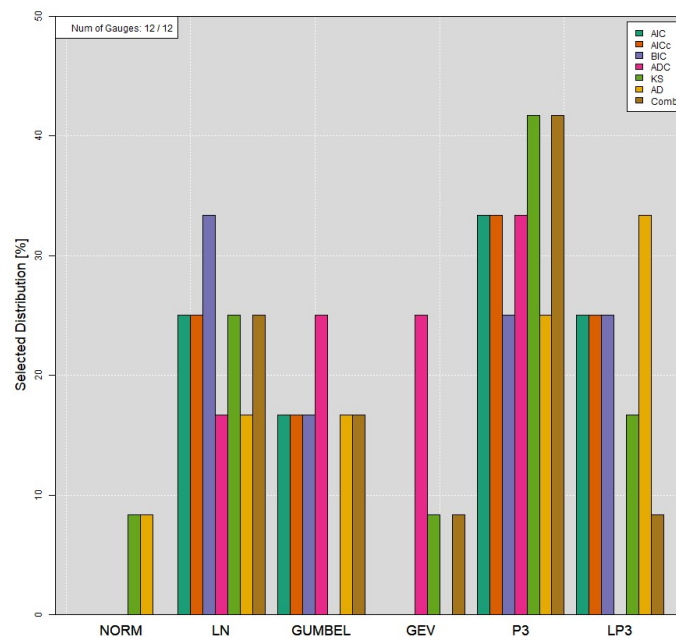


Figure 1.17 Distribution fit analysis for all considered 13 gauging station using various GoF criteria when the Method of Moments was used to estimate the parameters of the various distributions.

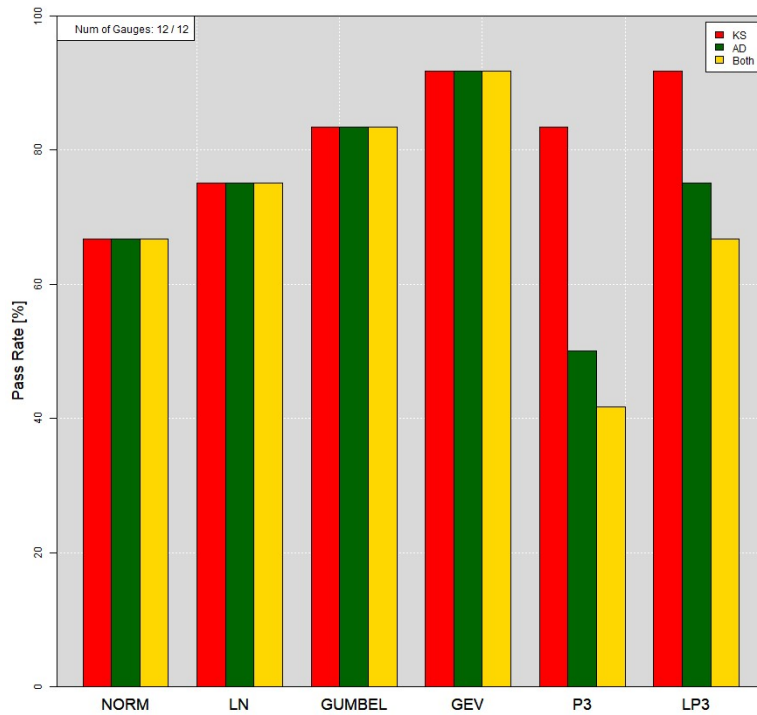


Figure 1.18 GoF passing rate for all considered 13 gauging station using GoF criteria such as KS, AD and combination of these two when the Maximum Likelihood Method was used to estimate the parameters of the various distributions.

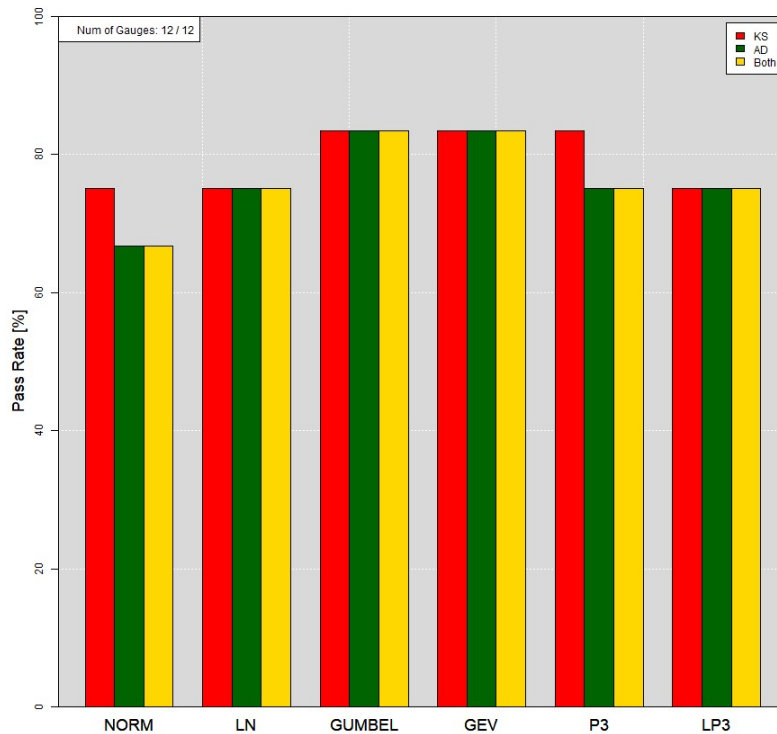


Figure 1.19 GoF passing rate for all considered 13 gauging station using GoF criteria such as KS, AD and combination of these two when the Method of Moments was used to estimate the parameters of the various distributions.

1.4.3 Probability Distribution Selection

When the Maximum Likelihood Estimation (MLE) and Method of Moments (MOM) approaches were employed for parameter estimation of various probability distributions, the most suitable probability distribution for the gauge stations in question was ranked based on the goodness of fit (GoF) metrics and are provided in Tables 1.5 and 1.6.

The ranking of the best fitted probability distribution for the considered gauging stations based on goodness of fit (GoF) criteria presented in Table 1.5 and 1.6, when the MLE and MOM methods were applied to estimate parameter of these distributions, respectively.

Table 1.5 Probability Distribution Selection (Maximum Likelihood Method)

Station name	Gauge ID	Rank_1	Rank_2	Rank_3	Rank_4	Rank_5	Rank_6
Karad	10001	P3	LN	LP3	GUMBEL	GEV	NORM
Warunji	10002	LN	GEV	LP3	P3	GUMBEL	NORM
Samdoli (Seasonal)	10004	P3	LN	GUMBEL	GEV	NORM	LP3
Arjunwad (Seasonal)	10005	LN	P3	GUMBEL	GEV	LP3	NORM
Terwad (Seasonal)	10006	GUMBEL	GEV	LN	P3	LP3	NORM
Kurundwad	10007	GUMBEL	LN	LP3	GEV	P3	NORM
Sadalga (Seasonal)	10008	P3	LN	GUMBEL	LP3	GEV	NORM
Gokak Falls (Seasonal)	10009	GUMBEL	P3	GEV	LP3	NORM	LN
Mudhol	10010	GEV	LN	LP3	GUMBEL	P3	NORM
Koyna	10011	LP3	LN	P3	GUMBEL	GEV	NORM
Karad	10001	LN	P3	LP3	GEV	GUMBEL	NORM
Warunji	10002	LN	GEV	LP3	P3	GUMBEL	NORM
Samdoli (Seasonal)	10004	LP3	GEV	P3	GUMBEL	LN	NORM
Arjunwad (Seasonal)	10005	LN	LP3	P3	GUMBEL	GEV	NORM
Terwad (Seasonal)	10006	GUMBEL	GEV	LN	P3	LP3	NORM
Kurundwad	10007	GUMBEL	LN	GEV	LP3	P3	NORM
Sadalga (Seasonal)	10008	LN	GUMBEL	LP3	P3	GEV	NORM
Gokak Falls (Seasonal)	10009	P3	GEV	LP3	GUMBEL	NORM	LN
Mudhol	10010	GEV	LP3	LN	P3	GUMBEL	NORM
Koyna	10011	P3	LP3	LN	GEV	GUMBEL	NORM
Sangali (Irwin Bridge)	10012	P3	LN	GEV	LP3	GUMBEL	NORM
Rajaram KT weir	10013	P3	LP3	NORM	LN	GUMBEL	NA

Table 1.6 Probability Distribution Selection (Method of Moments)

Station name	Gauge ID	Rank_1	Rank_2	Rank_3	Rank_4	Rank_5	Rank_6
Karad	10001	P3	LN	LP3	GUMBEL	GEV	NORM
Warunji	10002	LN	GEV	LP3	P3	GUMBEL	NORM
Samdoli (Seasonal)	10004	P3	LN	GUMBEL	GEV	NORM	LP3
Arjunwad (Seasonal)	10005	LN	P3	GUMBEL	GEV	LP3	NORM
Terwad (Seasonal)	10006	GUMBEL	GEV	LN	P3	LP3	NORM
Kurundwad	10007	GUMBEL	LN	LP3	GEV	P3	NORM
Sadalga (Seasonal)	10008	P3	LN	GUMBEL	LP3	GEV	NORM
Gokak Falls (Seasonal)	10009	GUMBEL	P3	GEV	LP3	NORM	LN
Mudhol	10010	GEV	LN	LP3	GUMBEL	P3	NORM
Koyna	10011	P3	LN	LP3	GUMBEL	GEV	NORM
Sangali (Irwin Bridge)	10012	P3	LN	LP3	GEV	GUMBEL	NORM
Rajaram KT weir	10013	P3	NORM	LP3	LN	GUMBEL	NA

1.4.4 Flood Frequency Analysis Results

The flood frequency analysis results for each gauging site are provided in the following sections:

The detailed flood frequency analysis (FFA) for considered gauging sites in UKB has been carried out using various probability distributions with use of various plotting positions methods in R programming software. The Goodness-of-Fit (GoF) criteria has been used to select various probability distribution at considered each gauging site. Alternatively, the flood frequency analysis results were estimated using Flood Frequency Distribution (FFD 2.1) MATLAB software. Further The FFD 2.1 results are used to cross verify the results achieved using the FFA in R programming.

Flood Frequency Analysis: Karad

According to the FFA for the Karad gauging site, the three best-fitting probability distributions are the Pearson type III, Log Normal, and Log Pearson III distributions. The probability plot derived using the best fit Pearson III distribution (MOM) for the Karad gauging site is displayed in Figure 1.18. Additionally, the flood quantiles for different return periods are summarised in this chart. When comparing the specified distribution curve to the observed data, the residual plot indicates the inaccuracy in the discharge value calculation. Confidence in the choice of flood quantile for different return periods is thus given. In general, this figure offers significant information about the possible flood levels anticipated for the Karad gauge location for different recurrence intervals.

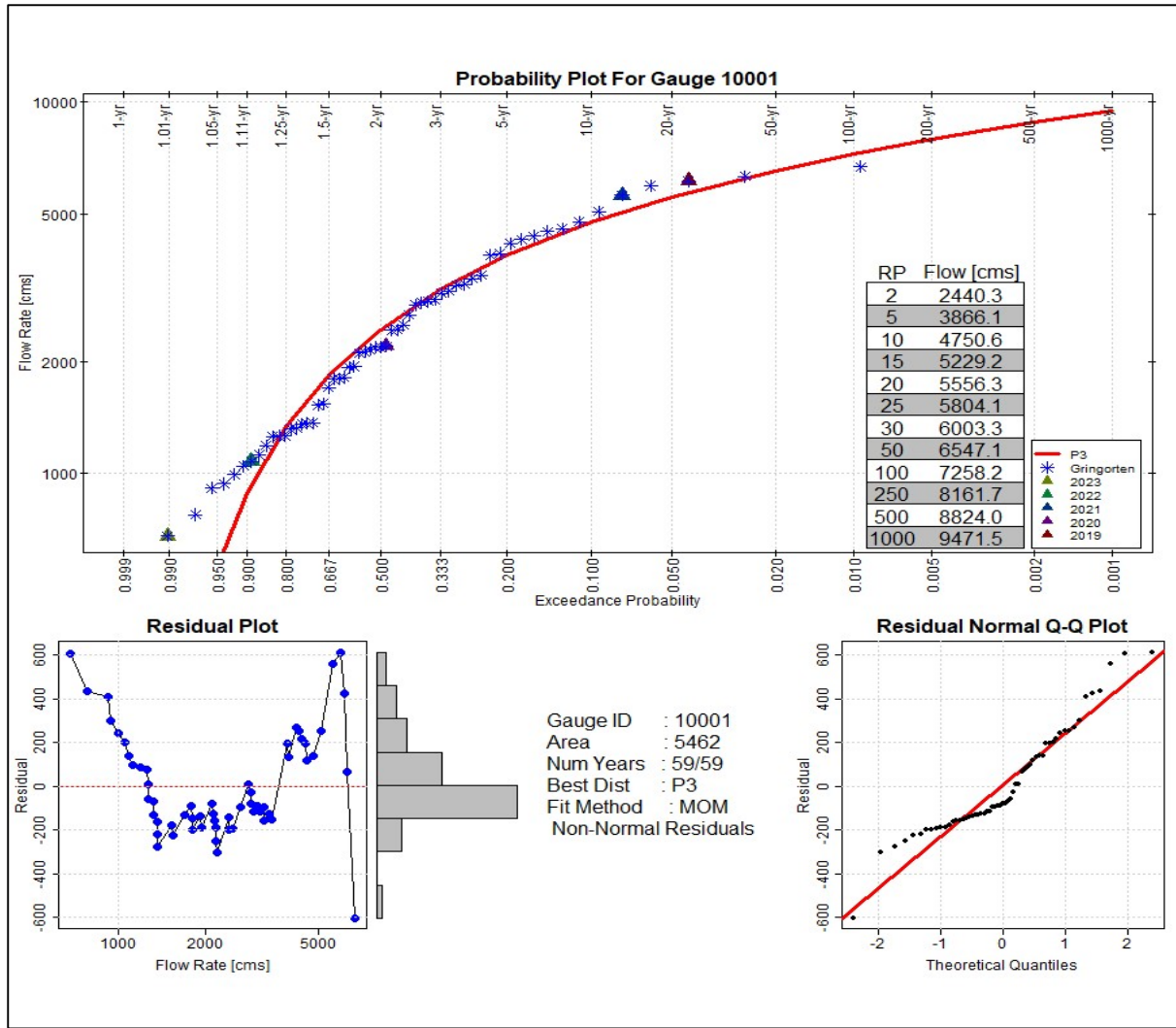


Figure 1.18 Best fit probability distribution results based on FFA approach for Karad.

From the FFD 2.1 result (Figure 1.19), it can be observed that several probability distributions can be accepted based on various performance metrics used for selection of these probability distributions which includes coefficient of determination (R^2), Root Mean Square Error (RMSE), Kolmogorov-Smirnov (KS) and Anderson Darling (AD) criterion. The best fitting distributions are Pearson III, General Extreme Value (GEV), Log Normal (2P), and Log Pearson 3 Distribution for the Karad gauging site. The results of this analysis for Karad gauging site are summarised in Figure 1.19 and Table 1.7. Among these distributions, the Pearson III stands out as the best-fitting distribution. This indicates that the Pearson III distribution provides the most accurate representation of the observed data among the tested distributions.

The flood frequency analysis results for Karad gauging site obtained with FFD 2.1 and FFA approach are comparable. Therefore, the Pearson III distribution (MOM) obtained with FFA approach (Figure 1.18) is recommended as the best fitted probability distribution for estimating flood quantiles for different return periods at Karad gauging site.

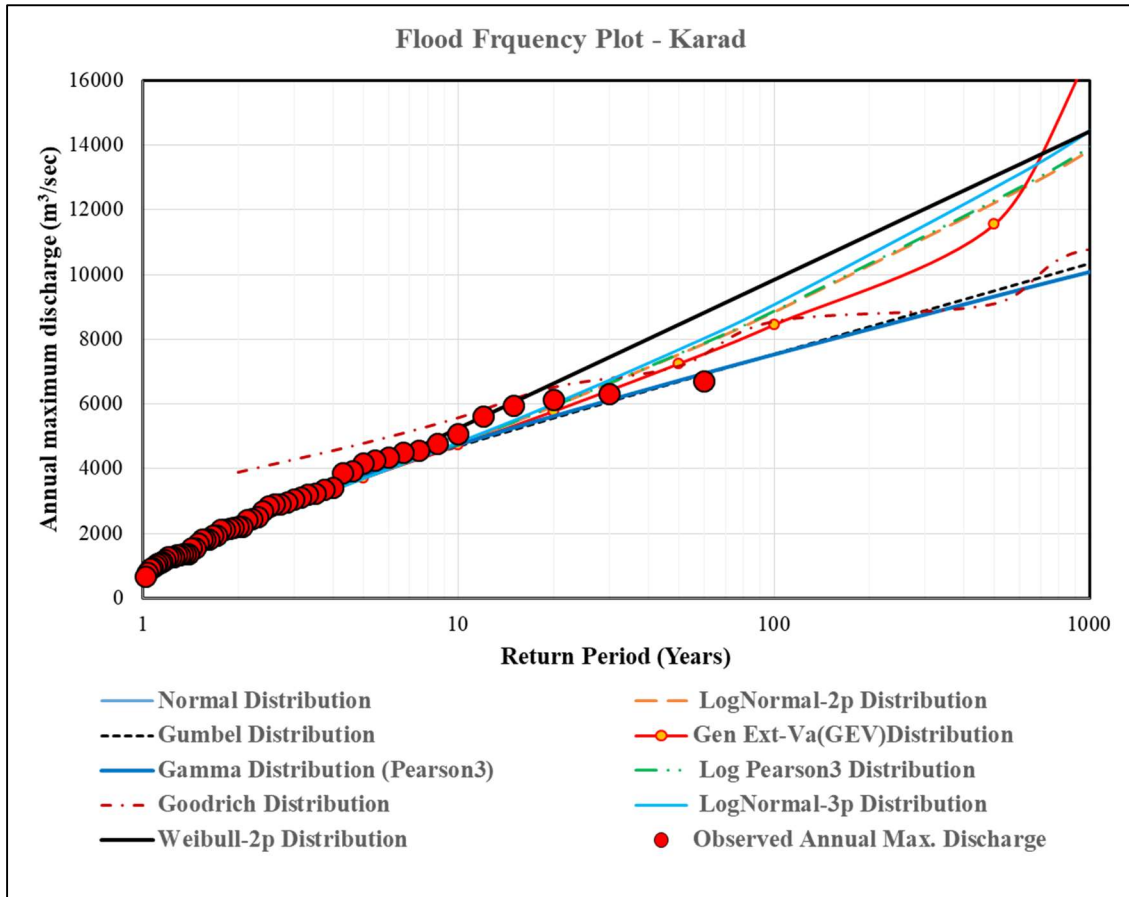


Figure 1.19 Flood frequency plot using FFD 2.1 software for Karad.

Table 1.7 Results of the flood frequency analysis using FFD 2.1 software at Karad

Return Period (Years)	Non-Exceedance probability	Gamma (Pearson3)	Gen Ext-Va (GEV)	LogNormal-2p	Log Pearson3 Distribution
		Quantile (%)	Quantile (%)	Quantile (%)	Quantile (%)
2	0.5	2372.6	2317.1	2262.4	2245.8
5	0.8	3814.5	3713.8	3705	3703.3
10	0.9	4753.7	4729.3	4794.7	4822.9
20	0.95	5631	5777.6	5932.5	6004.5
50	0.98	6732	7251.2	7539	7690.7
100	0.99	7534.6	8448.9	8845	9074
500	0.998	9331.2	11557.6	12222.3	12692.7
1000	0.999	10084	16900.1	13839.8	14443.4
10000	0.9999	12523.7	38944.4	20006.7	21197.7

Flood Frequency Analysis: Warunji

According to the FFA for the Warunji gauging site, the three best-fitting probability distributions are the Log Normal, Generalized Extreme Value and Log Pearson III distributions. The probability plot derived using the best fit Log Normal distribution (MOM) with parameter estimation method of moment for the Karad gauging site is displayed in Figure 1.20. Additionally, the flood quantiles for different return periods are summarised in this chart. When comparing the specified distribution curve to the observed data, the residual plot indicates the inaccuracy in the discharge value calculation. Confidence in the choice of flood quantile for different return periods is thus given. In general, this figure offers significant information about the possible flood levels anticipated for the Warunji gauge location for different recurrence intervals.

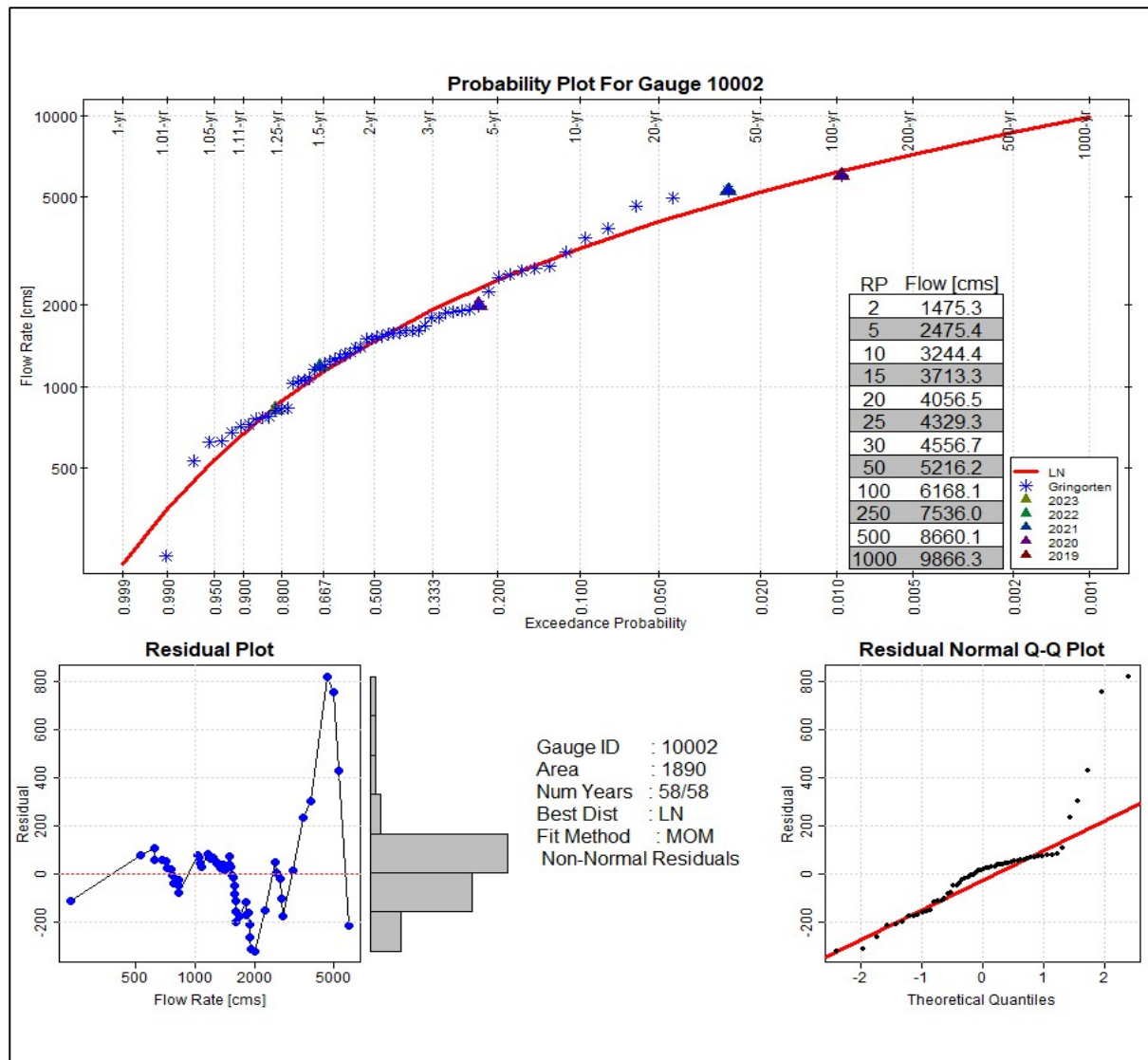


Figure 1.20 Best fit probability distribution results based on FFA approach for Warunji.

From the FFD 2.1 result (Figure 1.21), it can be observed that several probability distributions can be accepted based on various performance metrics used for selection of these probability distributions. The best fitting distributions are Log Normal (2P), Log Normal (3P) and Pearson III distributions for the Warunji gauging site. The results of this analysis for Warunji gauging site are summarised in Figure 1.21 and Table 1.8. Among these distributions, the Log Normal (2P) stands out as the best-fitting distribution. This indicates that the Log Normal (2P) distribution provides the most accurate representation of the observed data among the tested distributions.

The flood frequency analysis results for Warunji gauging site obtained with FFD 2.1 and FFA approach are comparable. Therefore, the Log Normal distribution (MOM) obtained with FFA approach (Figure 1.20) is recommended as the best fitted probability distribution for estimating flood quantiles for different return periods at Warunji gauging site.

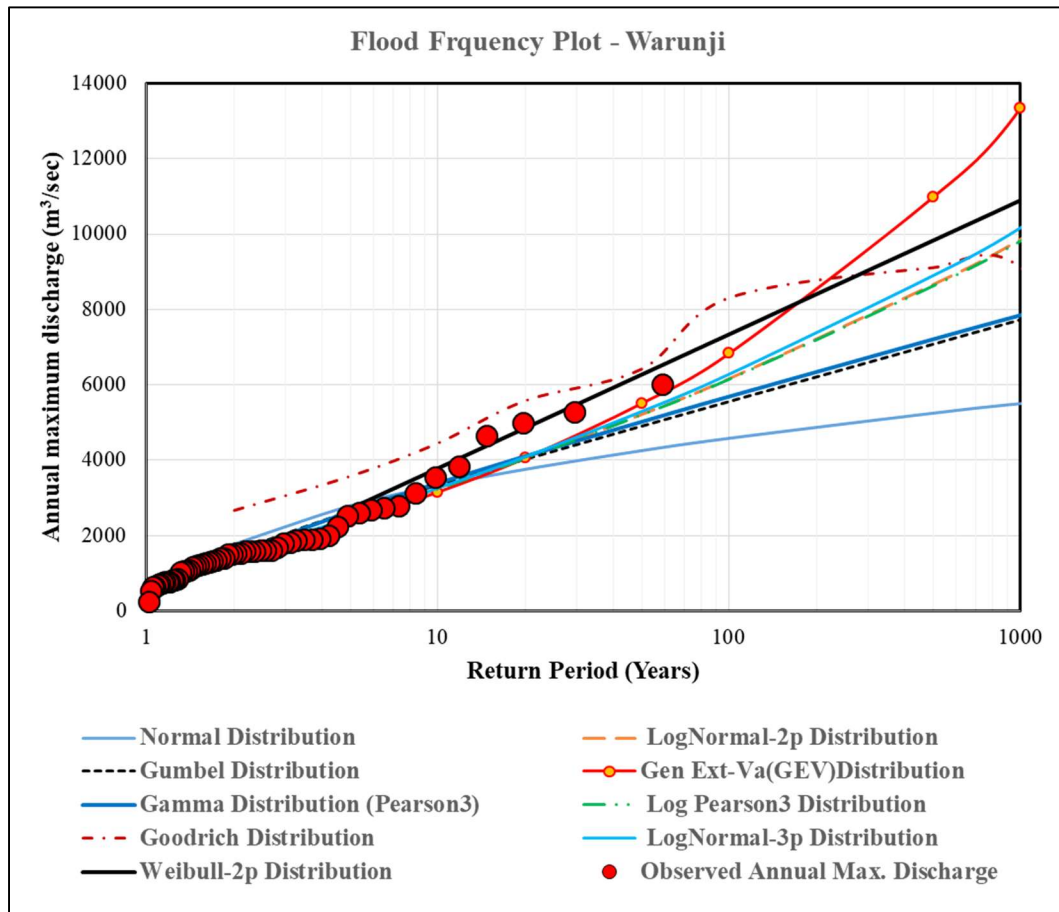


Figure 1.21 Flood frequency plot using FFD 2.1 software for Warunji.

Table 1.8 Results of the flood frequency analysis using FFD 2.1 software at Warunji

Return Period (Years)	Non-Exceedance probability	LogNormal-2p	LogNormal- 3p	Gamma Distribution (pearson3)
		Quantile (%)	Quantile (%)	Quantile (%)
2	0.5	1475.3	1468.6	1515.3
5	0.8	2475.4	2478.2	2637
10	0.9	3244.4	3262.7	3392.4
20	0.95	4056.5	4097	4109.5
50	0.98	5216.2	5296.2	5021.1
100	0.99	6168.1	6286.2	5692
500	0.998	8660.1	8896.9	7208.5
1000	0.999	9866.3	10168.7	7848.8
10000	0.9999	14523.7	15116	9938.4

Flood Frequency Analysis: Samdoli (Seasonal)

According to the FFA for the Samdoli gauging site, the three best-fitting probability distributions are the Pearson type III, Log Normal and Gumbel distributions. The probability plot derived using the best fit Pearson type III distribution (MLE) with Maximum Likelihood Method (MLE) of parameter estimation for the Samdoli gauging site is displayed in Figure 1.22. Additionally, the flood quantiles for different return periods are summarised in this chart. When comparing the specified distribution curve to the observed data, the residual plot indicates the inaccuracy in the discharge value calculation. Confidence in the choice of flood quantile for different return periods is thus given. In general, this figure offers significant information about the possible flood levels anticipated for the Samdoli gauge location for different recurrence intervals.

From the FFD 2.1 result (Figure 1.23), it can be observed that several probability distributions can be accepted based on various performance metrics used for selection of these probability distributions. The best fitting distributions are Normal Distribution, Gamma (Pearson III) and Gumbel distributions for the Samdoli gauging site. The results of this analysis for Samdoli gauging site are summarised in Figure 1.23 and Table 1.9. Among these distributions, the Normal distribution stands out as the best-fitting distribution. This indicates that the Normal distribution provides the most accurate representation of the observed data among the tested distributions.

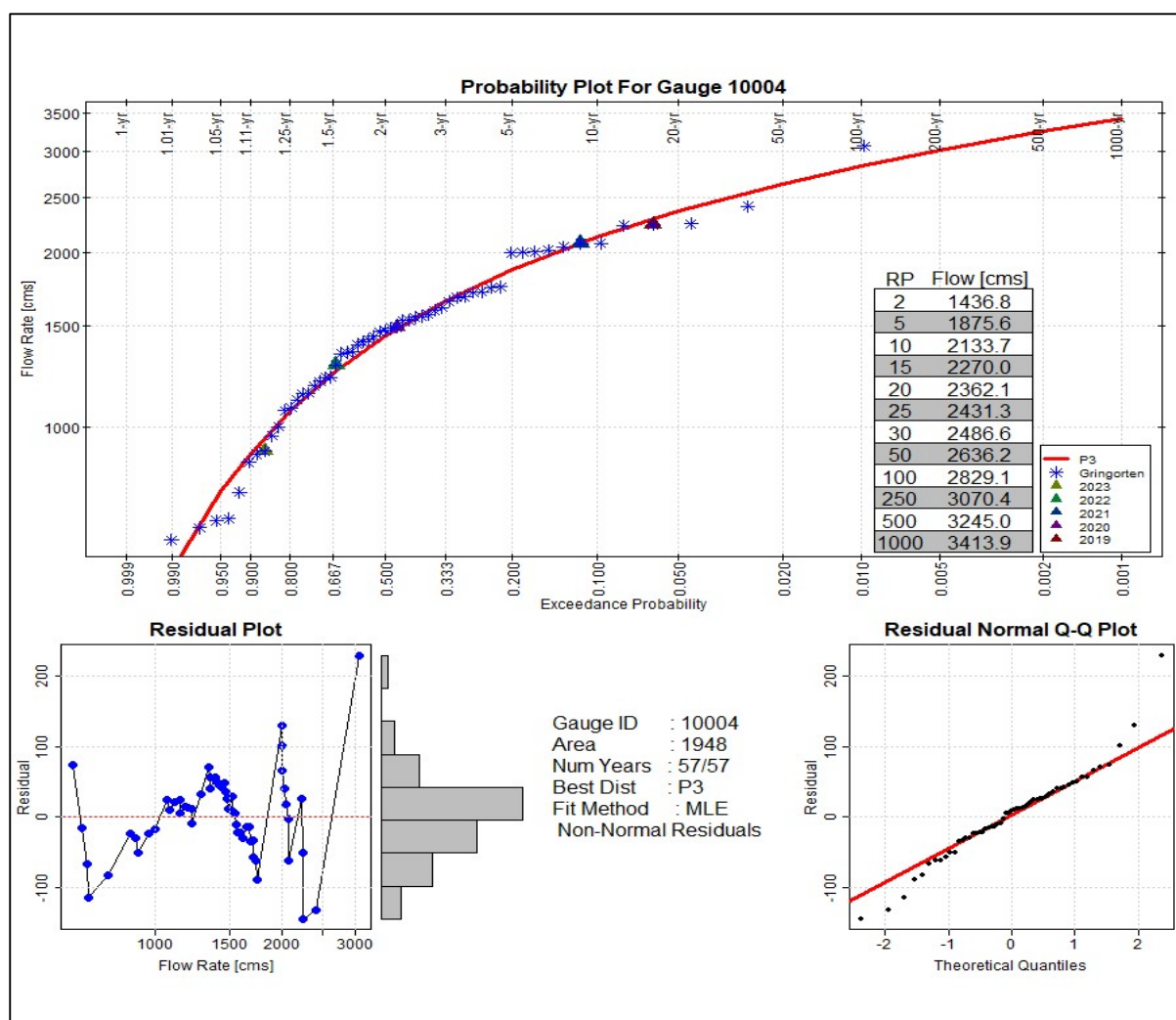


Figure 1.22. Best fit probability distribution results based on FFA approach for Samdoli.

The flood frequency analysis results for Samdoli gauging site obtained with FFD 2.1 and FFA approach are comparable. Therefore, the Pearson III distribution (MLE) obtained with FFA approach (Figure 1.22) is recommended as the best fitted probability distribution for estimating flood quantiles for different return periods at Samdoli gauging site.

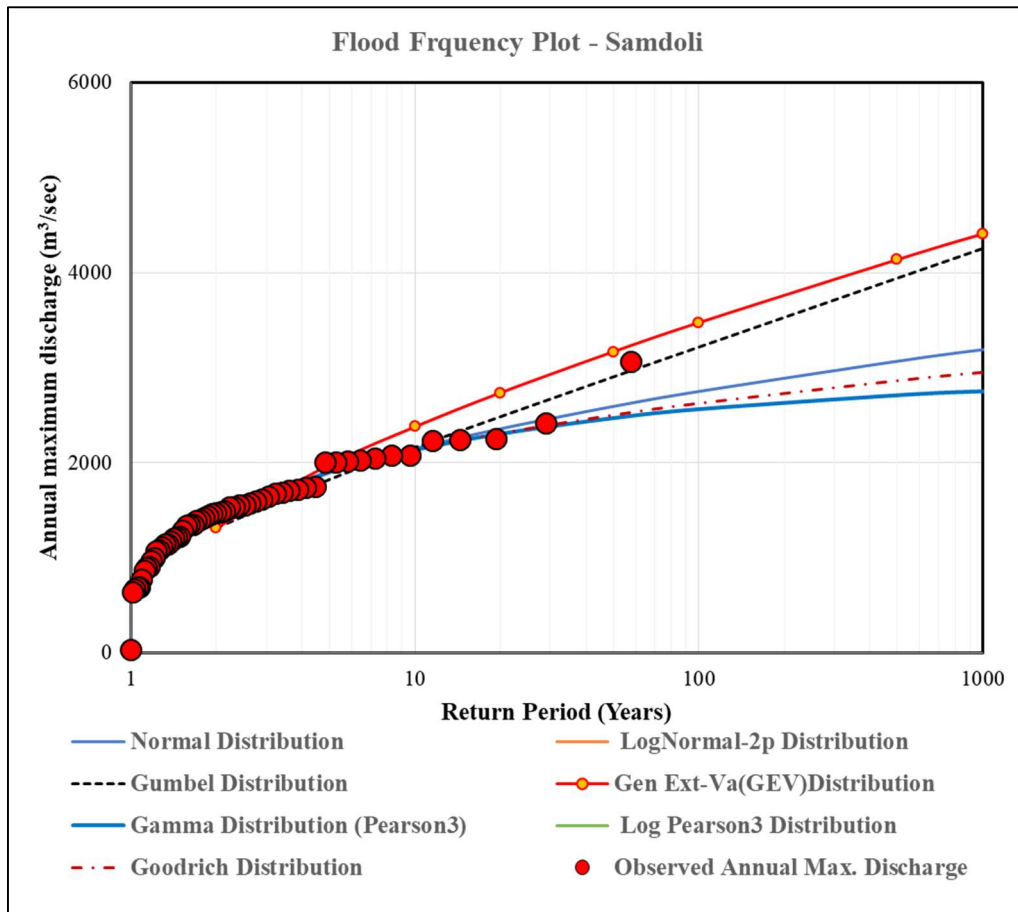


Figure 1.23 Flood frequency plot using FFD 2.1 software for Samdoli.

Table 1.9 Results of the flood frequency analysis using FFD 2.1 software at Samdoli

Return Period (Years)	Non-Exceedence probability	Normal Distribution	Gamma Distribution	Gumbel Distribution
		Quantile (%)	Quantile (%)	Quantile (%)
2	0.5	1411.7	1429.4	1317.1
5	0.8	1896.6	1908.6	1826.2
10	0.9	2150	2137	2163.3
20	0.95	2359.3	2306.8	2486.6
50	0.98	2594.9	2472.4	2905.2
100	0.99	2751.9	2565.9	3218.8
500	0.998	3069.8	2712.3	3943.5
1000	0.999	3192	2754.1	4255.1
10000	0.9999	3554.3	2838.9	5258.1

Flood Frequency Analysis: Arjunwad (Seasonal)

According to the FFA for the Samdoli gauging site, the three best-fitting probability distributions are the Log Normal, Pearson type III and Gumbel distributions. The probability plot derived using the best fit Log Normal distribution (MOM) with parameter estimation method of moment (MOM) for the Arjunwad gauging site is displayed in Figure 1.24. Additionally, the flood quantiles for different return periods are summarised in this chart. When comparing the specified distribution curve to the observed data, the residual plot indicates the inaccuracy in the discharge value calculation. Confidence in the choice of flood quantile for different return periods is thus given. In general, this figure offers significant information about the possible flood levels anticipated for the Arjunwad gauge location for different recurrence intervals.

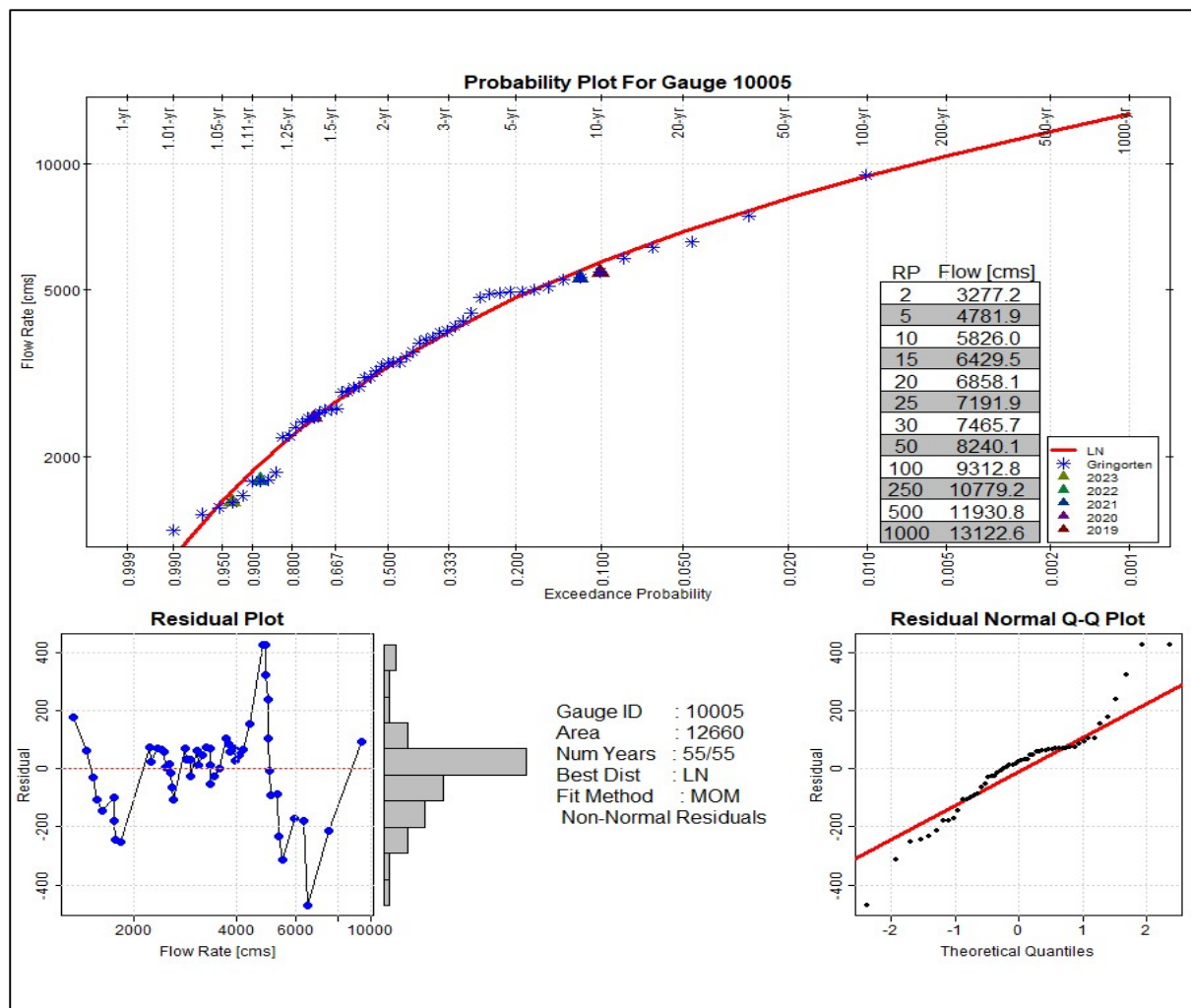


Figure 1.24 Best fit probability distribution results based on FFA approach for Arjunwad.

From the FFD 2.1 result (Figure 1.25), it can be observed that several probability distributions can be accepted based on various performance metrics used for selection of these probability distributions. The best fitting distributions are Log Normal 2P, Log Normal 3P and Log Pearson III distributions for the Arjunwad gauging site. The results of this analysis for Arjunwad gauging site are summarised in Figure 1.25 and Table 1.10. Among these distributions, the Log Normal 2P distribution stands out as the best-fitting distribution. This indicates that the Log Normal 2P distribution provides the most accurate representation of the observed data among the tested distributions.

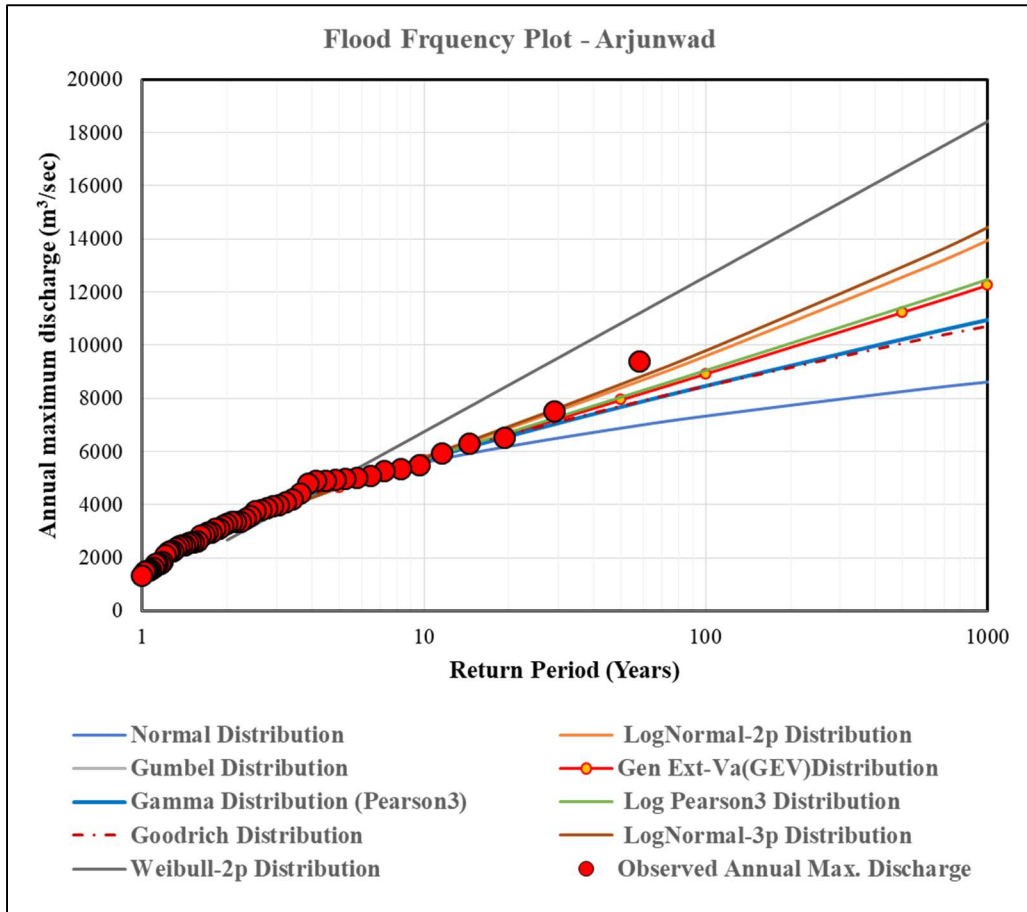


Figure 1.25. Flood Frequency Analysis: Arjunwad Station (Performance Metrics)

Table 1.10 Results of the flood frequency analysis using FFD 2.1 software at Arjunwad

Return Period (Years)	Non-Exceedence probability	Log Normal 2P	Log Normal 2P	Log Pearson 3 Distribution
		Quantile (%)	Quantile (%)	Quantile (%)
2	0.5	3073.8	3058.1	3114.6
5	0.8	4640.7	4642.9	4656.1
10	0.9	5755.7	5786.7	5703.5
20	0.95	6875.8	6946.2	6718.9
50	0.98	8399.3	8537.3	8047.4
100	0.99	9598.2	9799	9055.1
500	0.998	12574.4	12961.3	11433.6
1000	0.999	13949.7	14435.1	12481.1
10000	0.9999	18976.9	19875.5	16082.9

The flood frequency analysis results for Arjunwad gauging site obtained with FFD 2.1 and FFA approach are comparable. Therefore, the Log Normal distribution (MOM) obtained with FFA approach (Figure 1.24) is recommended as the best fitted probability distribution for estimating flood quantiles for different return periods at Arjunwad gauging site.

Flood Frequency Analysis: Terwad (Seasonal)

According to the FFA for the Terwad gauging site, the three best-fitting probability distributions are the Gumbel, Generalized Extreme Value (GEV), Log Normal distributions. The probability plot derived using the best fit Gumbel distribution (MOM) with parameter estimation method of moment (MOM) for the Terwad gauging site is displayed in Figure 1.26. Additionally, the flood quantiles for different return periods are summarised in this chart. When comparing the specified distribution curve to the observed data, the residual plot indicates the inaccuracy in the discharge value calculation. Confidence in the choice of flood quantile for different return periods is thus given. In general, this figure offers significant information about the possible flood levels anticipated for the Terwad gauge location for different recurrence intervals.

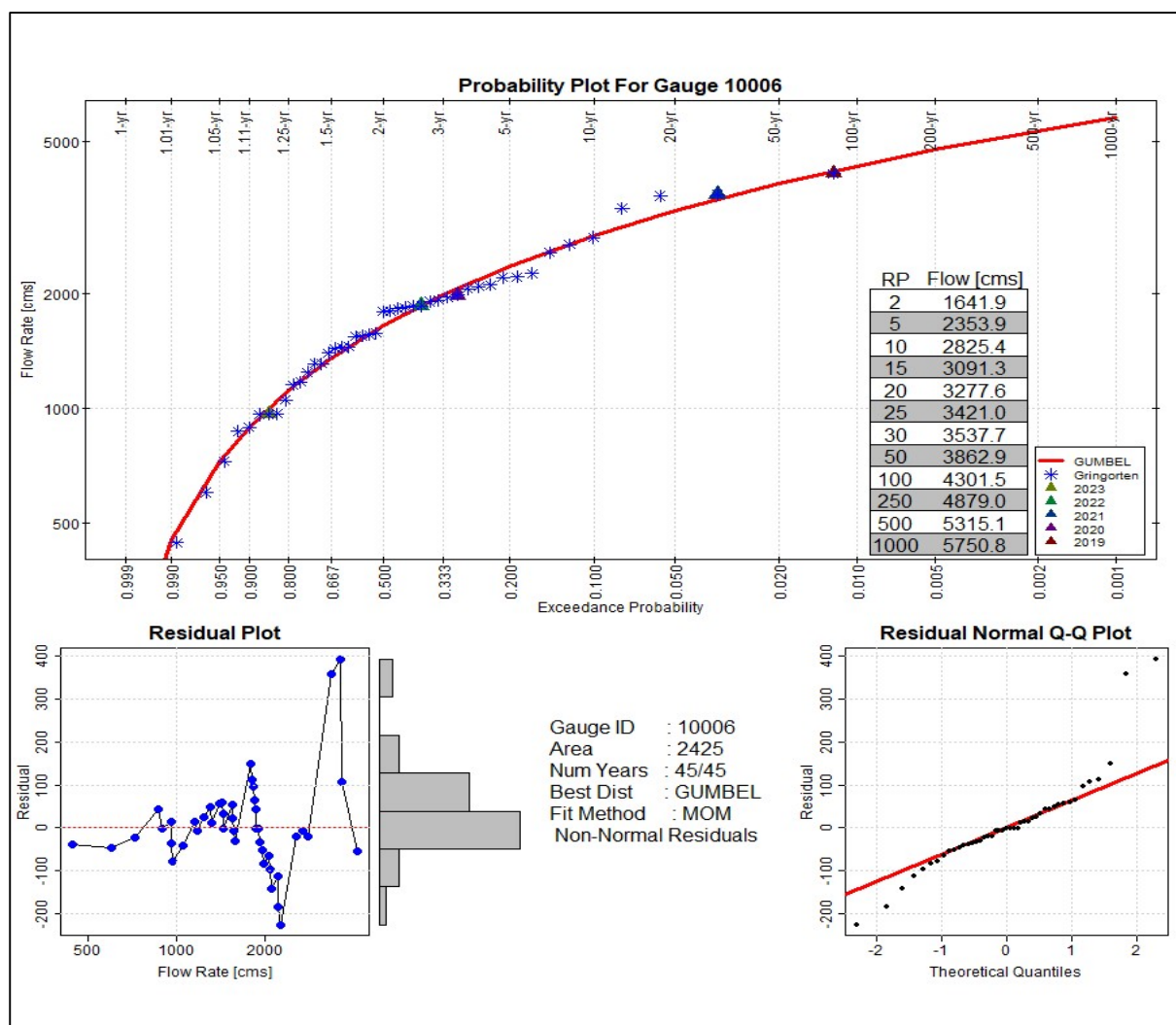


Figure 1.26 Best fit probability distribution results based on FFA approach for Terwad.

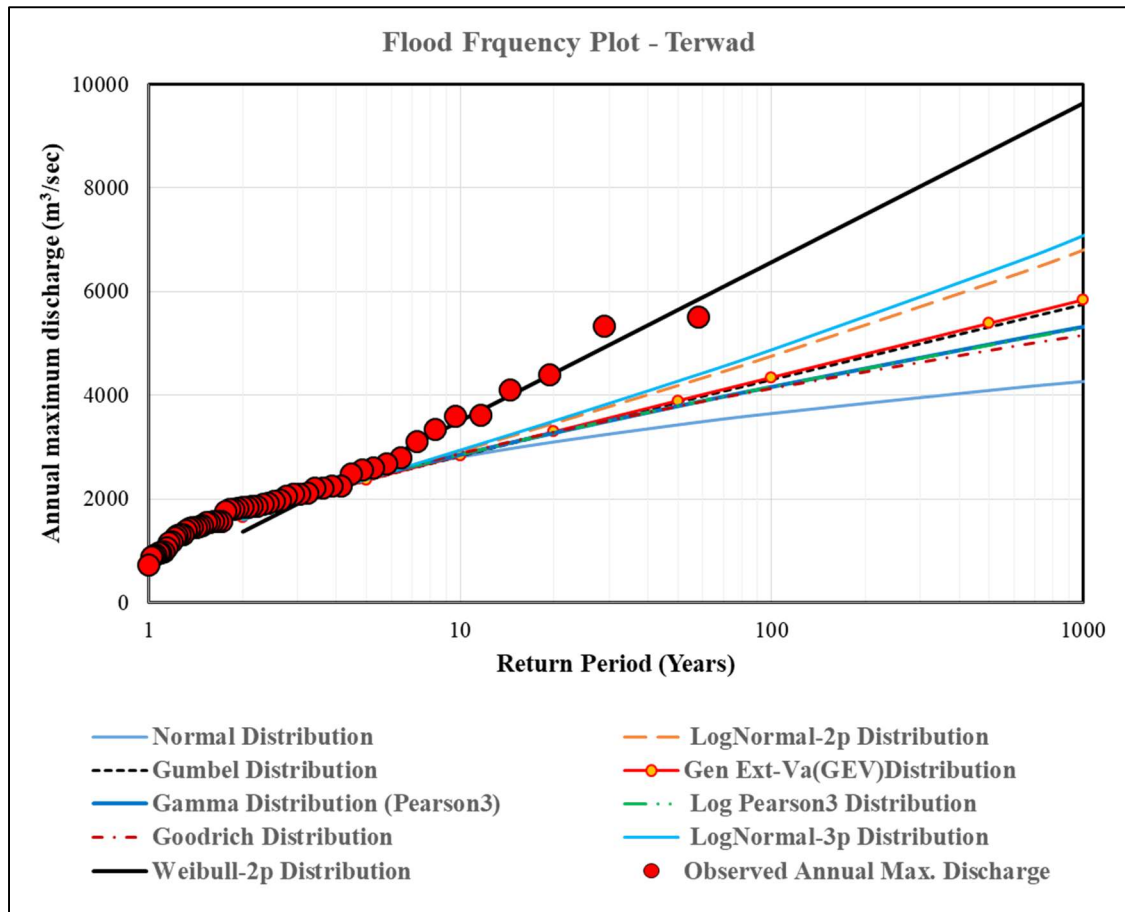


Figure 1.27 Flood frequency plot using FFD 2.1 software for Terwad.

Table 1.11 Results of the flood frequency analysis using FFD 2.1 software at Terwad

Return Period (Years)	Non-Excedence probability	Weibull-2p Distribution	Gamma (Pearson3)	Gen Ext-Va (GEV)
		Quantile (%)	Quantile (%)	Quantile (%)
2	0.5	1366.2	1653.9	1638.6
5	0.8	2584.8	2393.1	2358
10	0.9	3506.7	2853.2	2836.3
20	0.95	4428.6	3273	3296.5
50	0.98	5647.3	3790	3894.4
100	0.99	6569.1	4161.4	4343.9
500	0.998	8709.7	4980.2	5387.9
1000	0.999	9631.6	5319.2	5839
10000	0.9999	12694	6405.2	7346.2

From the FFD 2.1 result (Figure 1.27), it can be observed that several probability distributions can be accepted based on various performance metrics used for selection of these probability distributions. The best fitting distributions are Weibull (2P), Gamma (Pearson 3) and General Extreme Value (GEV) distributions for the Terwad gauging site. The results of this analysis for Terwad gauging site are summarised in Figure 1.27 and Table 1.11. Among these distributions, the Weibull (2P) distribution stands out as the best-fitting distribution. This indicates that the Weibull (2P) distribution provides the most accurate representation of the observed data among the tested distributions.

The flood frequency results for Terwad gauging site obtained with FFD 2.1 and FFA approach are comparable. Therefore, the Gumbel distribution (MOM) obtained with FFA approach (Figure 1.24) is recommended as the best fitted probability distribution for estimating flood quantiles for different return periods at Terwad gauging site.

Flood Frequency Analysis: Kurundwad

According to the FFA for the Kurundwad gauging site, the three best-fitting probability distributions are the Gumbel, Log Normal and Pearson type III distributions. The probability plot derived using the best fit Gumbel distribution (MOM) for the Kurundwad gauging site is displayed in Figure 1.28. Additionally, the flood quantiles for different return periods are

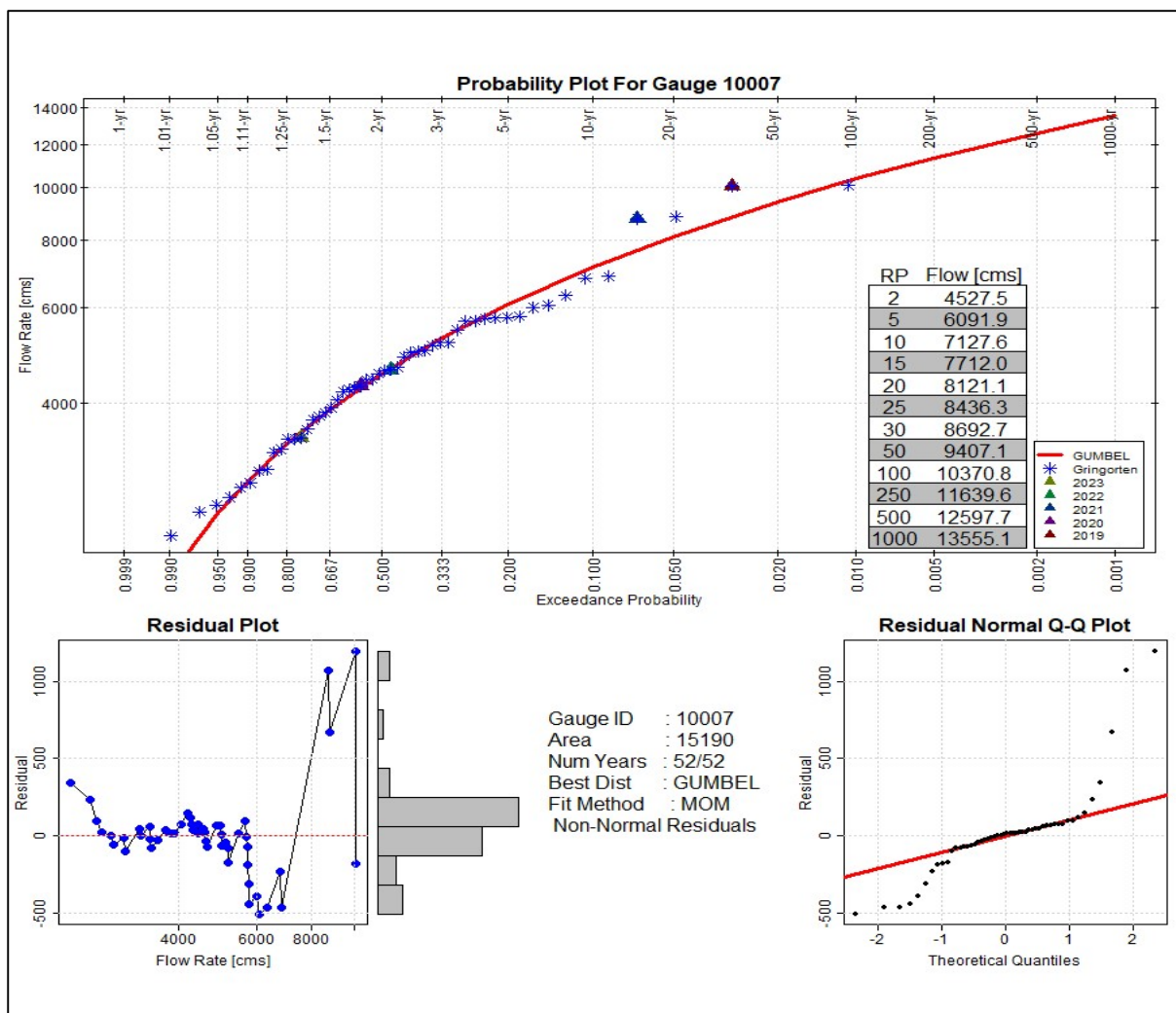


Figure 1.28 Best fit probability distribution results based on FFA approach for Kurundwad.

summarised in this chart. When comparing the specified distribution curve to the observed data, the residual plot indicates the inaccuracy in the discharge value calculation. Confidence in the choice of flood quantile for different return periods is thus given. In general, this figure offers significant information about the possible flood levels anticipated for the Kurundwad gauge location for different recurrence intervals.

From the FFD 2.1 result (Figure 1.29), it can be observed that several probability distributions can be accepted based on various performance metrics used for selection of these probability distributions. The best fitting distributions are Pearson III, General Extreme Value (GEV) and Log Pearson 3 distributions for the Kurundwad gauging site. The results of this analysis for Kurundwad gauging site are summarised in Figure 1.29 and Table 1.12. Among these distributions, the Pearson III distribution stands out as the best-fitting distribution. This indicates that the Pearson III distribution provides the most accurate representation of the observed data among the tested distributions.

The flood frequency analysis results for Kurundwad gauging site obtained with FFD 2.1 and FFA approach are comparable. Therefore, the Gumbel distribution (MOM) obtained with FFA approach (Figure 1.28) is recommended as the best fitted probability distribution for estimating flood quantiles for different return periods at Kurundwad gauging site.

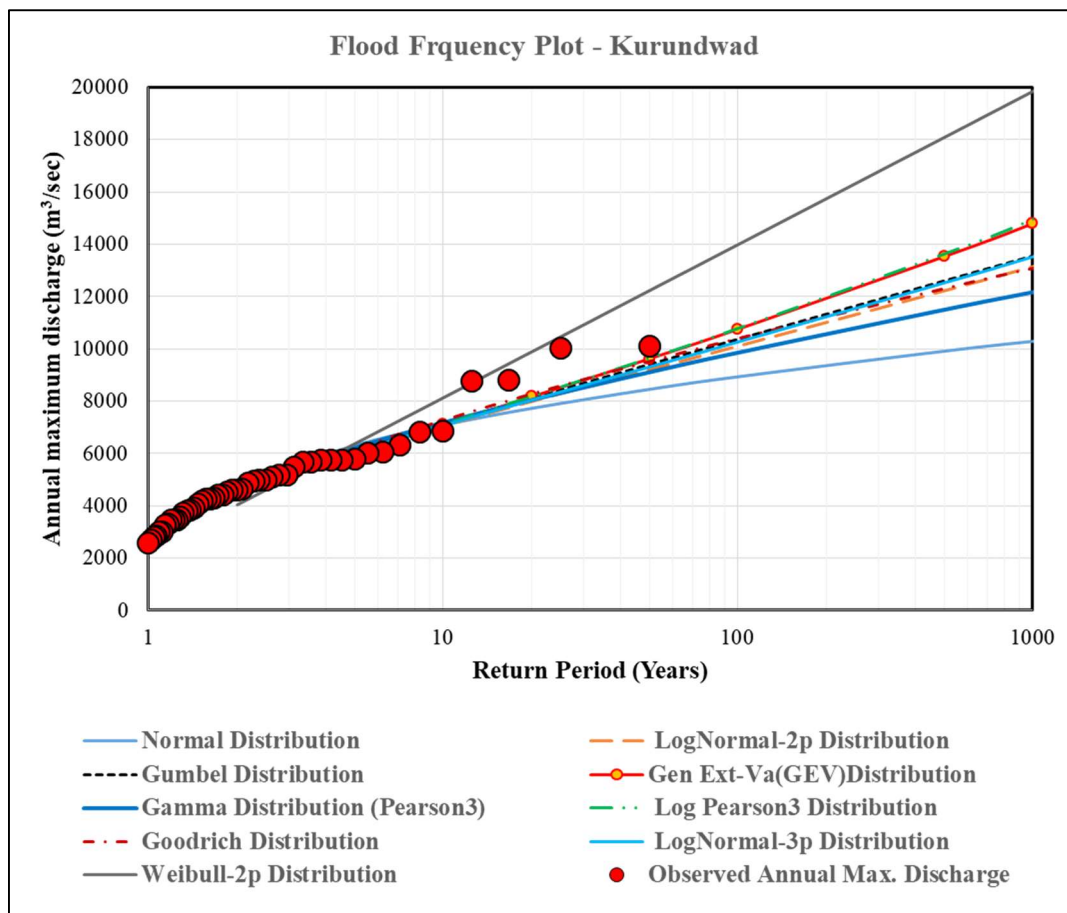


Figure 1.29 Flood frequency plot using FFD 2.1 software for Kurundwad.

Table 1.12 Results of the flood frequency analysis using FFD 2.1 software at Kurundwad

Return Period (Years)	Non-Exceedence probability	Gamma (Pearson3)	Gen Ext-Va (GEV)	Log Pearson 3
		Quantile (%)	Quantile (%)	Quantile (%)
2	0.5	4603.4	4479	4473
5	0.8	6210.7	6027.5	6032.6
10	0.9	7180.2	7103.5	7114.3
20	0.95	8050.5	8175.3	8189.5
50	0.98	9107.3	9622.9	9643.1
100	0.99	9858.3	10754	10784.9
500	0.998	11494.5	13528.5	13627.4
1000	0.999	12165.2	14793.8	14949.2
10000	0.9999	14293.8	19333.7	19846.7

Flood Frequency Analysis: Sadalga (Seasonal)

According to the FFA for the Sadalga gauging site, the three best-fitting probability distributions are the Pearson type III, Log Normal, and Gumbel distributions. The probability plot derived using the best fit Pearson type III distribution (MOM) for the Sadalga gauging site is displayed in Figure 1.30. Additionally, the flood quantiles for different return periods are summarised in this chart. When comparing the specified distribution curve to the observed data, the residual plot indicates the inaccuracy in the discharge value calculation. Confidence in the choice of flood quantile for different return periods is thus given. In general, this figure offers significant information about the possible flood levels anticipated for the Sadalga gauge location for different recurrence intervals.

From the FFD 2.1 result (Figure 1.31), it can be observed that several probability distributions can be accepted based on various performance metrics used for selection of these probability distributions. The best fitting distributions are Pearson III, General Extreme Value (GEV) and Log Pearson 3 distributions for the Sadalga gauging site. The results of this analysis for Sadalga gauging site are summarised in Figure 1.31 and Table 1.13. Among these distributions, the Pearson III distribution stands out as the best-fitting distribution. This indicates that the Pearson III distribution provides the most accurate representation of the observed data among the tested distributions.

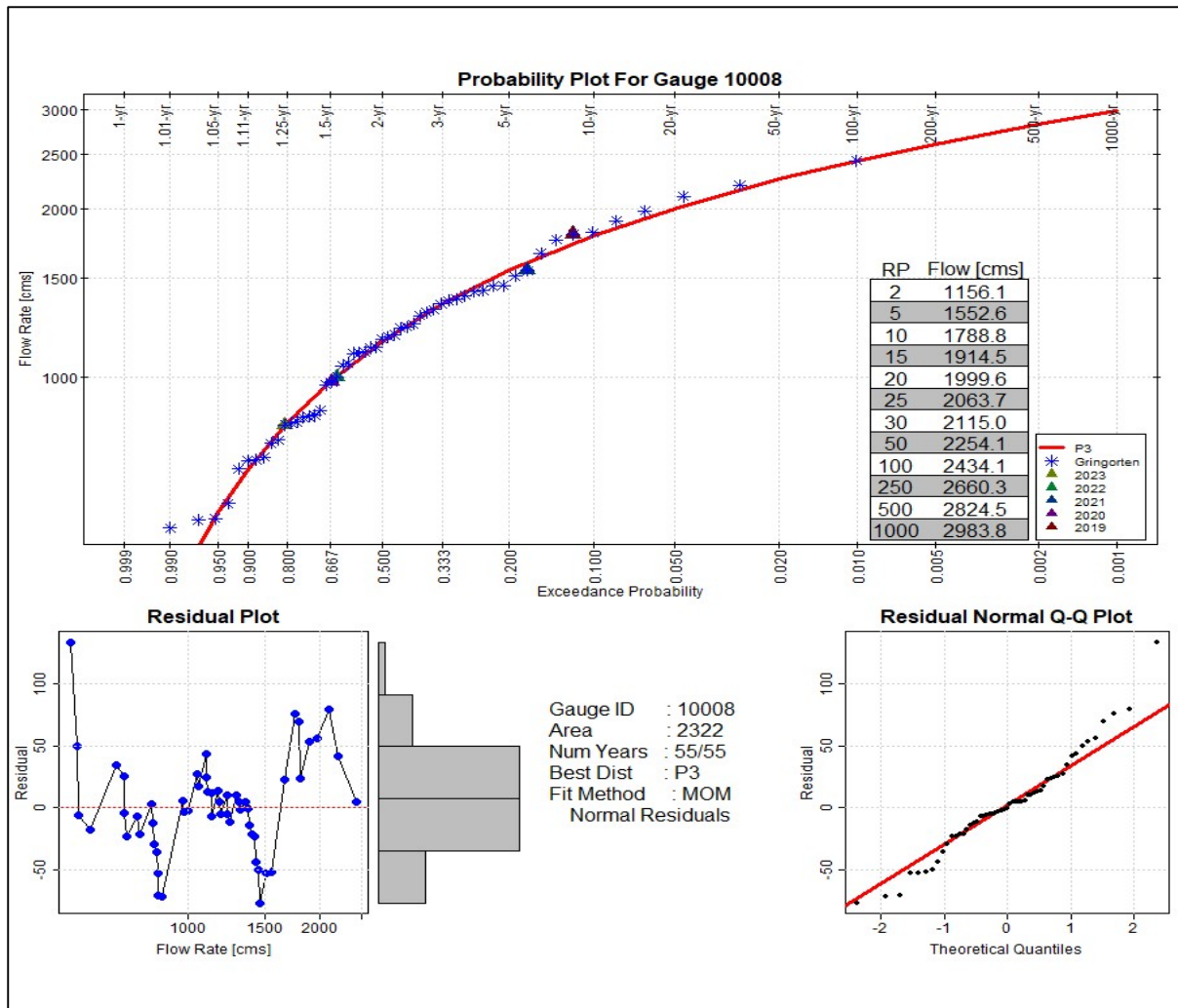


Figure 1.30 Best fit probability distribution results based on FFA approach for Sadalga.

The flood frequency analysis results for Sadalga gauging site obtained with FFD 2.1 and FFA approach are comparable. Therefore, the Pearson type III distribution (MOM) obtained with FFA approach (Figure 1.30) is recommended as the best fitted probability distribution for estimating flood quantiles for different return periods at Sadalga gauging site.

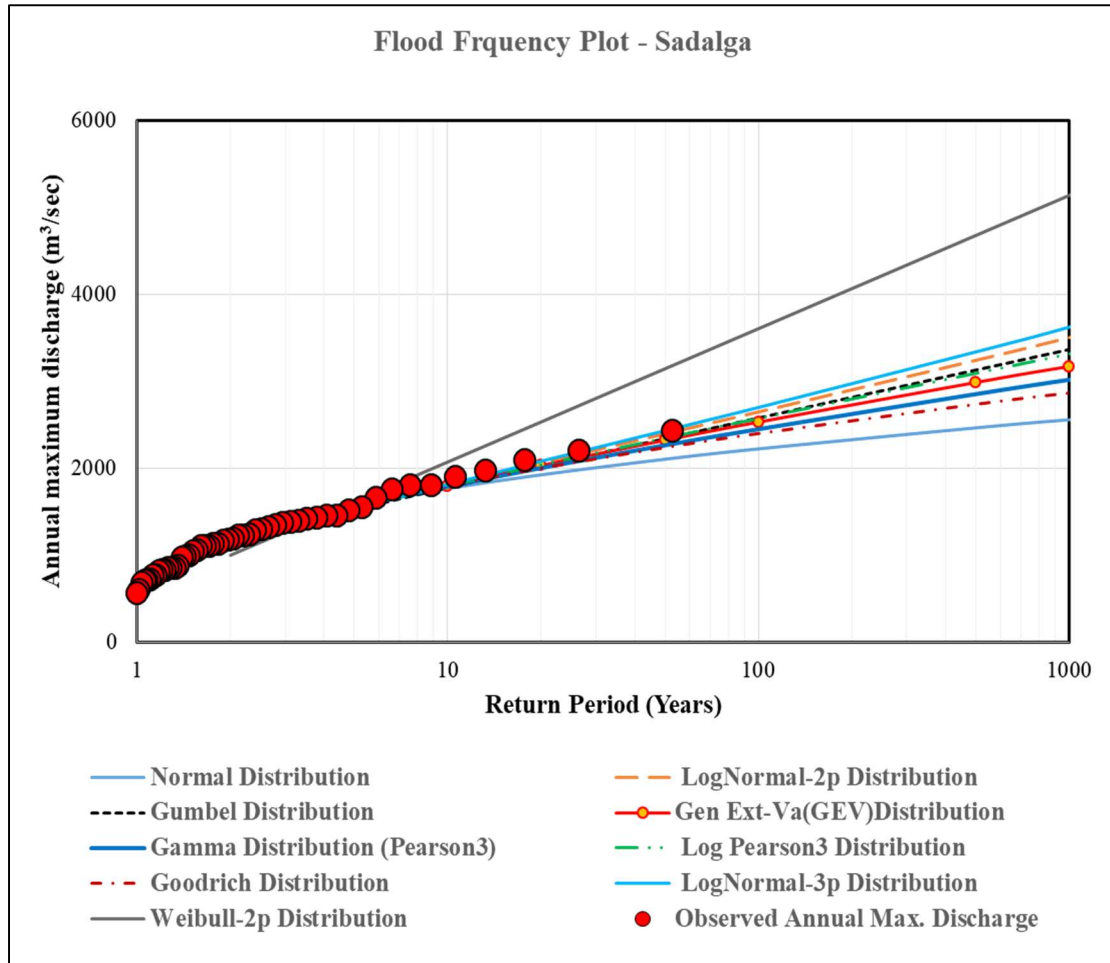


Figure 1.31 Flood frequency plot using FFD 2.1 software for Sadalga.

Table 1.13 Results of the flood frequency analysis using FFD 2.1 software at Sadalga

Return Period (Years)	Non-Exceedance probability	Gamma (Pearson3)	Gen Ext-Va (GEV)	Log Pearson 3
		Quantile (%)	Quantile (%)	Quantile (%)
2	0.5	1151.9	1143	1136.5
5	0.8	1550	1545.1	1540
10	0.9	1789.9	1798.9	1798.7
20	0.95	2005	2033.4	2040.9
50	0.98	2266.2	2324.4	2348
100	0.99	2451.7	2533.7	2575.1
500	0.998	2855.6	2989.7	3095.3
1000	0.999	3021.1	3174.5	3318.8
10000	0.9999	2836	3743.4	3546.2

Flood Frequency Analysis: Gokak Falls (Seasonal)

According to the FFA for the Gokak Falls gauging site, the three best-fitting probability distributions are the Gumbel, Pearson type III, and Generalized Extreme Value (GEV) distributions. The probability plot derived using the best fit Gumbel distribution (MOM) for the Gokak Falls gauging site is displayed in Figure 1.32. Additionally, the flood quantiles for different return periods are summarised in this chart. When comparing the specified distribution curve to the observed data, the residual plot indicates the inaccuracy in the discharge value calculation. Confidence in the choice of flood quantile for different return periods is thus given. In general, this figure offers significant information about the possible flood levels anticipated for the Gokak Falls gauge location for different recurrence intervals.

From the FFD 2.1 result (Figure 1.33), it can be observed that several probability distributions can be accepted based on various performance metrics used for selection of these probability distributions. The best fitting distributions are Gumbel, Gamma (Pearson III) and Weibull (2P) distributions for the Gokak Falls gauging site. The results of this analysis for Gokak Falls gauging site are summarised in Figure 1.33 and Table 1.14. Among these distributions, the Gumbel distribution stands out as the best-fitting distribution.

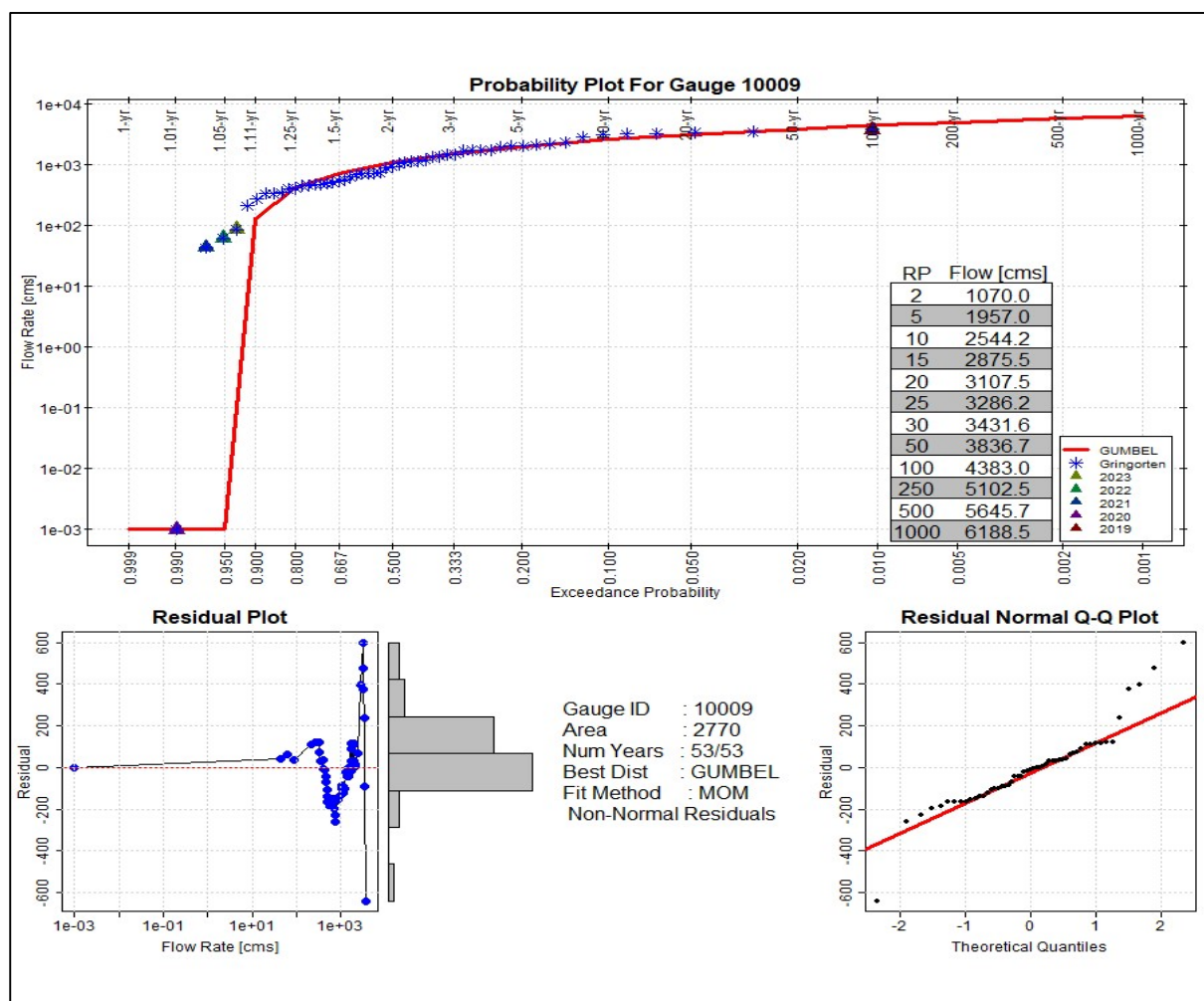


Figure 1.32 Best fit probability distribution results based on FFA approach for Gokak.

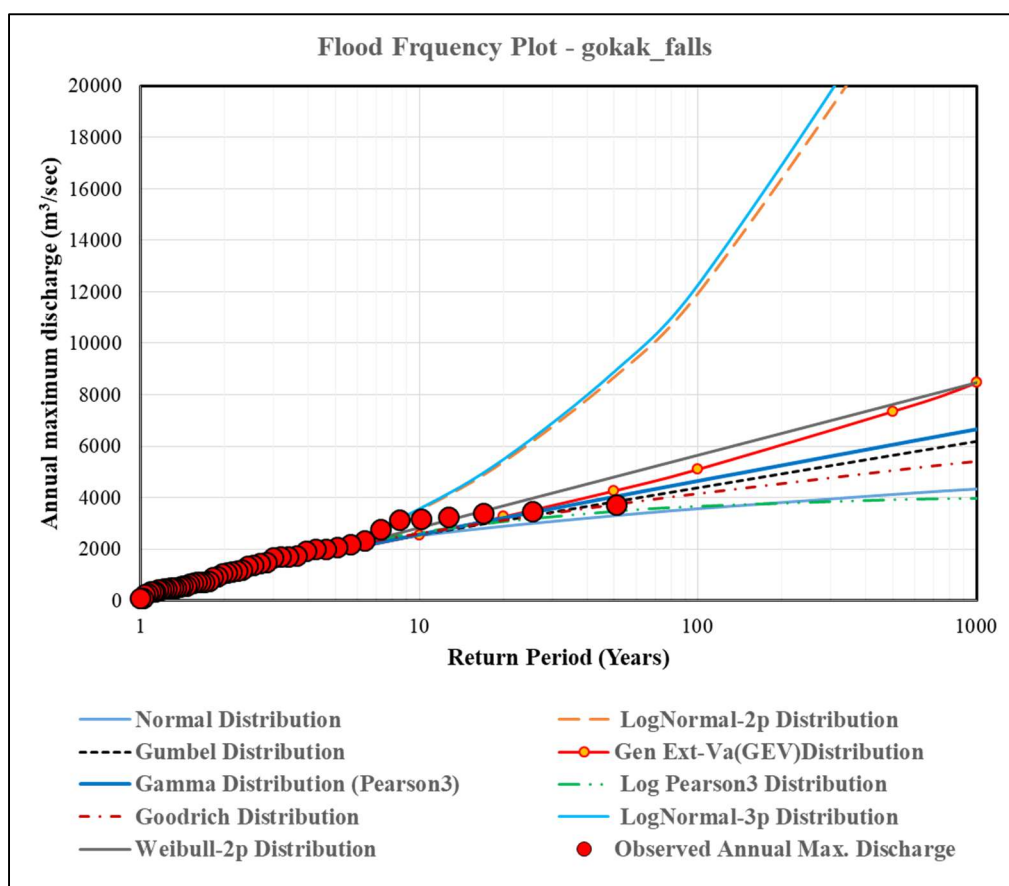


Figure 1.33 Flood frequency plot using FFD 2.1 software for Gokak Falls.

Table 1.14 Results of the flood frequency analysis using FFD 2.1 software at Gokak falls

Return Period (Years)	Non-Exceedance probability	Gumbel	Gamma (Pearson3)	Weibull-2p
		Quantile (%)	Quantile (%)	Quantile (%)
2	0.5	1070.2	976.5	859.1
5	0.8	1957	1908.8	1981.7
10	0.9	2544.1	2567.4	2830.8
20	0.95	3107.3	3206.6	3679.9
50	0.98	3836.3	4033.1	4802.5
100	0.99	4382.5	4648.9	5651.6
500	0.998	5644.9	6058.1	7623.3
1000	0.999	6187.5	6658.5	8472.4
10000	0.9999	7989.4	8634.6	11293.2

The flood frequency analysis results for Gokak Falls gauging site obtained with FFD 2.1 and FFA approach are comparable. Therefore, the Gumbel distribution (MOM) obtained with FFA approach (Figure 1.32) is recommended as the best fitted probability distribution for estimating flood quantiles for different return periods at Gokak Falls gauging site.

Flood Frequency Analysis: Mudhol

According to the FFA for the Mudhol gauging site, the three best-fitting probability distributions are the that Generalized Extreme Value, Log Normal, and Log Pearson type III distributions. The probability plot derived using the best fit Generalized Extreme Value (GEV) distribution (MLE) for the Mudhol gauging site is displayed in Figure 1.34. Additionally, the flood quantiles for different return periods are summarised in this chart. When comparing the specified distribution curve to the observed data, the residual plot indicates the inaccuracy in the discharge value calculation. Confidence in the choice of flood quantile for different return periods is thus given. In general, this figure offers significant information about the possible flood levels anticipated for the Mudhol gauge location for different recurrence intervals. However, most of the probability distribution method do not provide good fit for this gauging site so results may be used with cautions.

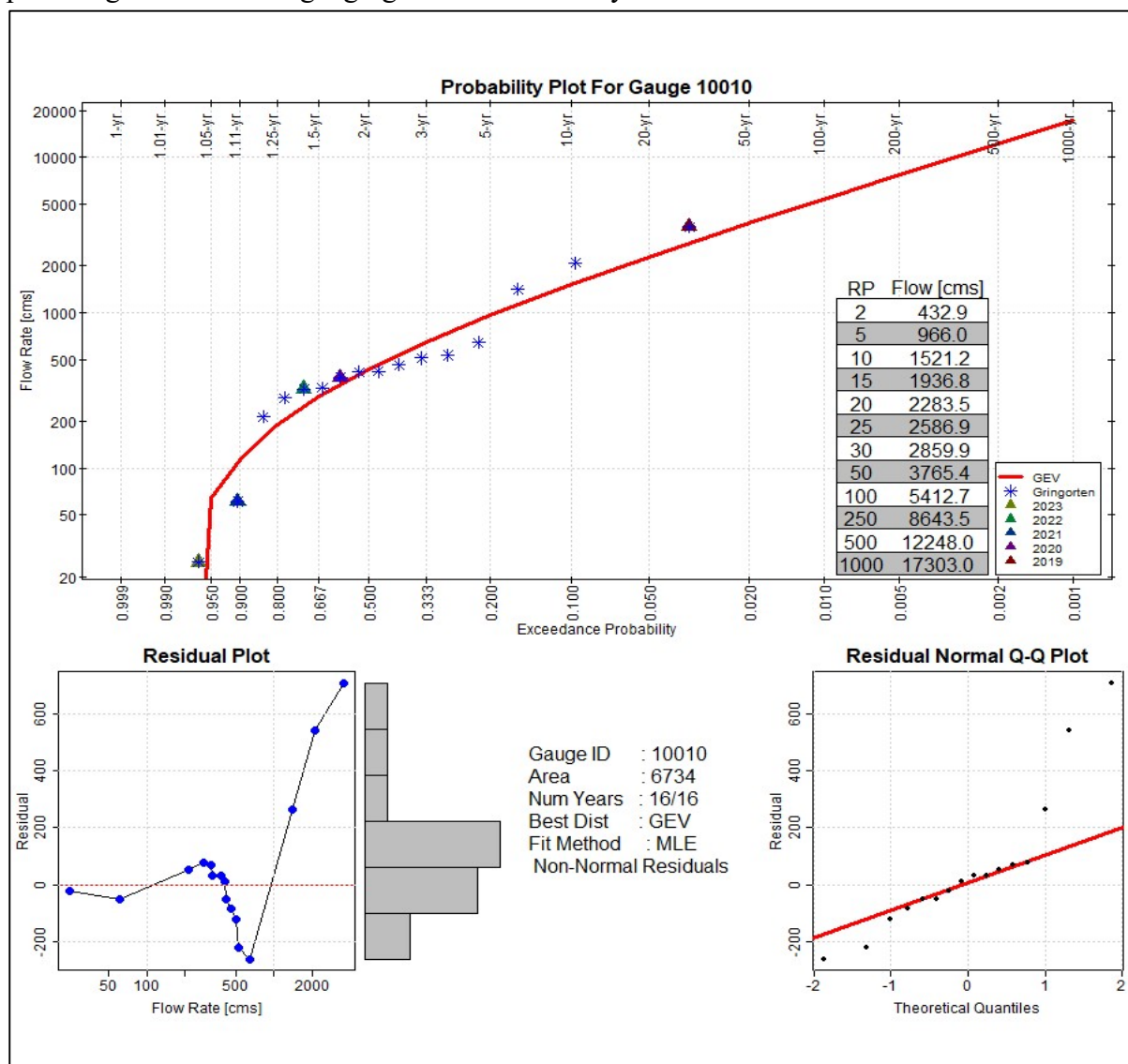


Figure 1.34 Best fit probability distribution results based on FFA approach for Mudhol.

From the FFD 2.1 result (Figure 1.35), it can be observed that several probability distributions can be accepted based on various performance metrics used for selection of these probability distributions. The best fitting distributions are GEV, Log Normal 2P and Log Normal 3P distributions for the Mudhol gauging site. The results of this analysis for Mudhol gauging site are summarised in Figure 1.35 and Table 1.15. Among these distributions, the GEV distribution stands out as the best-fitting distribution.

The flood frequency analysis results for Mudhol gauging site obtained with FFD 2.1 and FFA approach are comparable. Therefore, the GEV distribution (MLE) obtained with FFA approach (Figure 1.34) is recommended as the best fitted probability distribution for estimating flood quantiles for different return periods at Mudhol gauging site. However, as mentioned earlier as the most of the probability distributions do not provide good fit for this gauging site so results may be used with cautions.

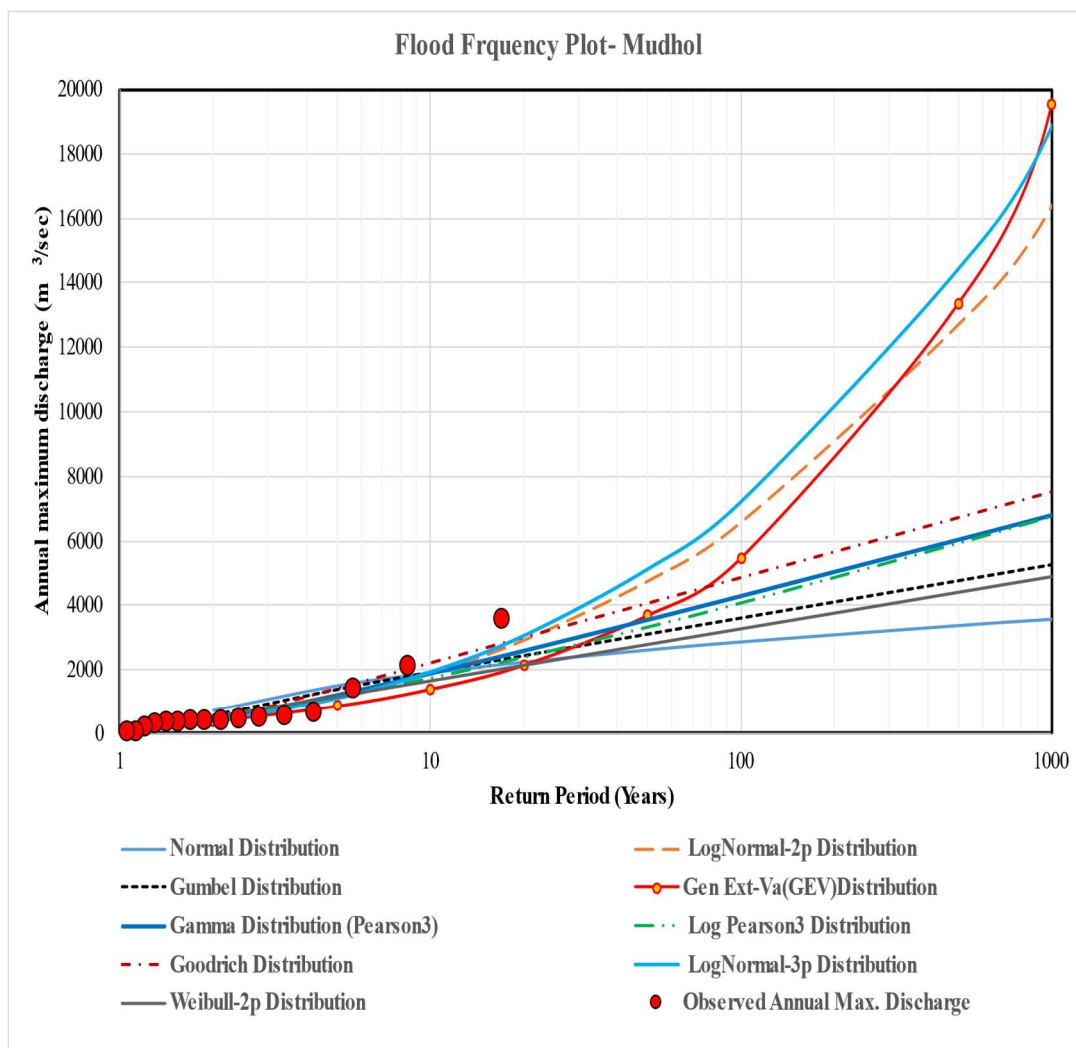


Figure 1.35 Flood frequency plot using FFD 2.1 software for Mudhol.

Table 1.15 Results of the flood frequency analysis using FFD 2.1 software at Mudhol

Return Period (Years)	Non-Exceedence probability	Gen Ext-Va (GEV)	LogNormal- 2p	LogNormal- 3p
		Quantile (%)	Quantile (%)	Quantile (%)
2	0.5	397.4	406.2	397.7
5	0.8	882.8	1111.1	1130.4
10	0.9	1412.9	1880.2	1957.3
20	0.95	2169.1	2903	3082.3
50	0.98	3705.3	4733.5	5141.2
100	0.99	5485.7	6557.4	7232.3
500	0.998	13370.6	12684.6	14435
1000	0.999	19535.2	16345.5	18827.5
10000	0.9999	68295.3	34666.4	41393.3

Flood Frequency Analysis: Koyna

According to the FFA for the Koyna gauging site, the three best-fitting probability distributions are the Pearson type III, Log Normal and Log Pearson type III distributions. The probability plot derived using the best fit Pearson type III distribution (MOM) for the Koyna gauging site is displayed in Figure 1.36. Additionally, the flood quantiles for different return periods are summarised in this chart. When comparing the specified distribution curve to the observed data, the residual plot indicates the inaccuracy in the discharge value calculation. Confidence in the choice of flood quantile for different return periods is thus given. In general, this figure offers significant information about the possible flood levels anticipated for the Koyna gauge location for different recurrence intervals.

From the FFD 2.1 result (Figure 1.37), it can be observed that several probability distributions can be accepted based on various performance metrics used for selection of these probability distributions. The best fitting distributions are Pearson type III, Log Pearson type III, and Log Normal distributions for the Koyna gauging site. The results of this analysis for Koyna gauging site are summarised in Figure 1.37 and Table 1.16. Among these distributions, the Pearson III distribution stands out as the best-fitting distribution.

The flood frequency analysis results for Koyna gauging site obtained with FFD 2.1 and FFA approach are comparable. Therefore, the Pearson III distribution (MOM) obtained with FFA approach (Figure 1.36) is recommended as the best fitted probability distribution for estimating flood quantiles for different return periods at Koyna gauging site.

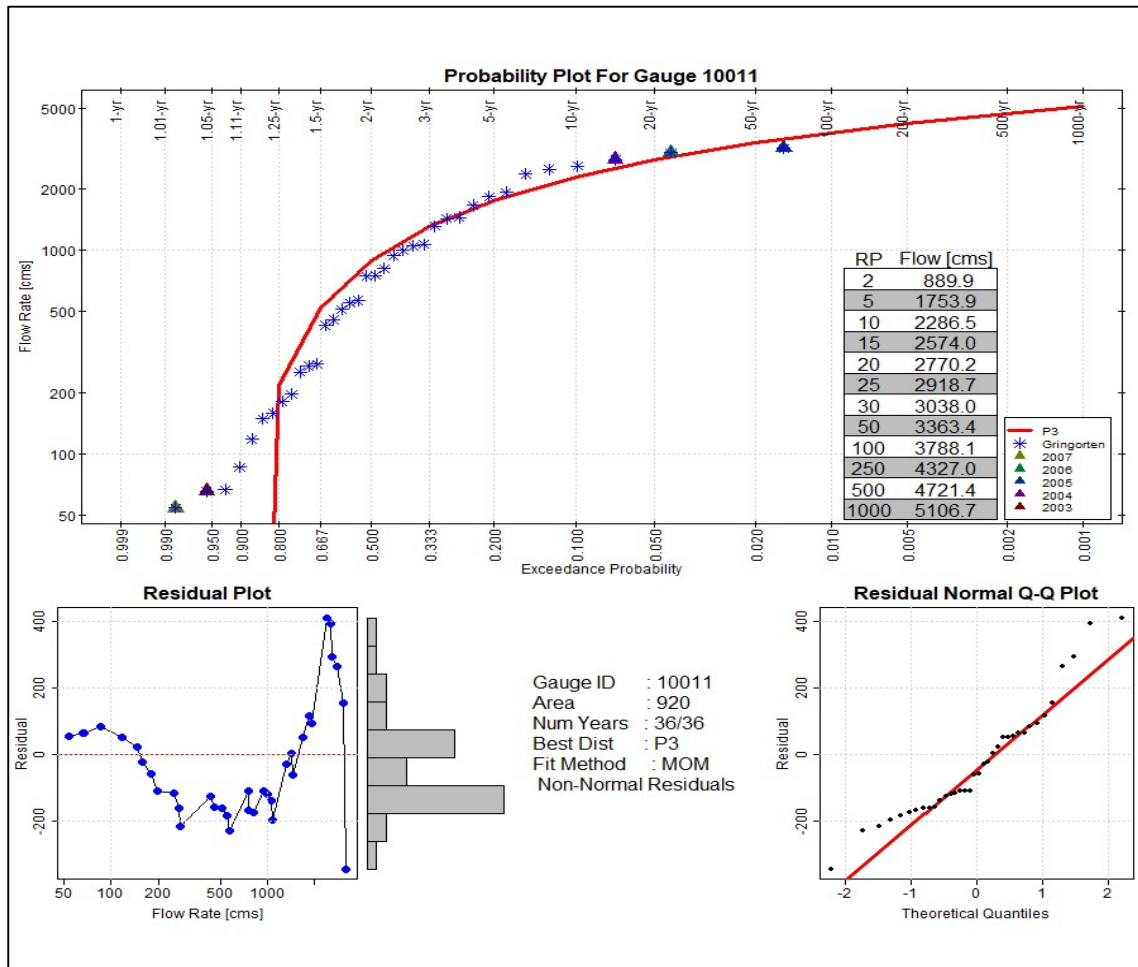


Figure 1.36 Best fit probability distribution results based on FFA approach for Koyna.

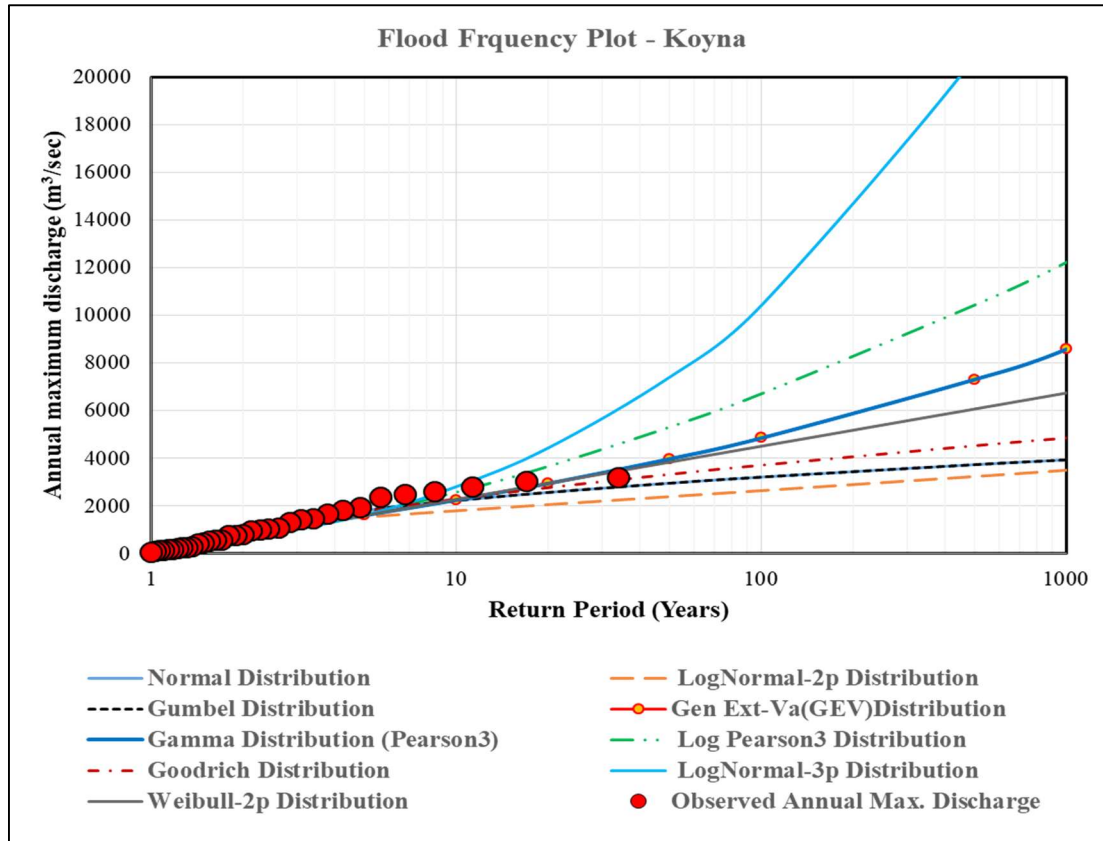


Figure 1.37 Flood frequency plot using FFD 2.1 software for Koyna.

Table 1.16 Results of the flood frequency analysis using FFD 2.1 software at Koyna

Return Period (Years)	Non-Exceedance probability	Gamma (Pearson3)	Log Normal 3p	Log Normal 2p
		Quantile (%)	Quantile (%)	Quantile (%)
2	0.5	777.4	634.6	1348.9
5	0.8	1612.4	1644.8	2241.9
10	0.9	2255.8	2585	2741.3
20	0.95	2950.8	3670.7	3163.8
50	0.98	3979.4	5318.7	3646.5
100	0.99	4858.4	6718.6	3971.2
500	0.998	7313	10440.2	4633.3
1000	0.999	8581.8	12230.6	4889
10000	0.9999	13978.2	18886.6	5650.1

Flood Frequency Analysis: Sangali (Irwin Bridge)

According to the FFA for the Sangali (Irwin Bridge) gauging site, the three best-fitting probability distributions are the Pearson type III, Log Normal and Log Pearson type III distributions. The probability plot derived using the best fit Pearson type III distribution (MOM) for the Sangali (Irwin Bridge) gauging site is displayed in Figure 1.38. Additionally, the flood quantiles for different return periods are summarised in this chart. When comparing the specified distribution curve to the observed data, the residual plot indicates the inaccuracy in the discharge value calculation. Confidence in the choice of flood quantile for different return periods is thus given. In general, this figure offers significant information about the possible flood levels anticipated for the Sangali (Irwin Bridge) gauge location for different recurrence intervals.

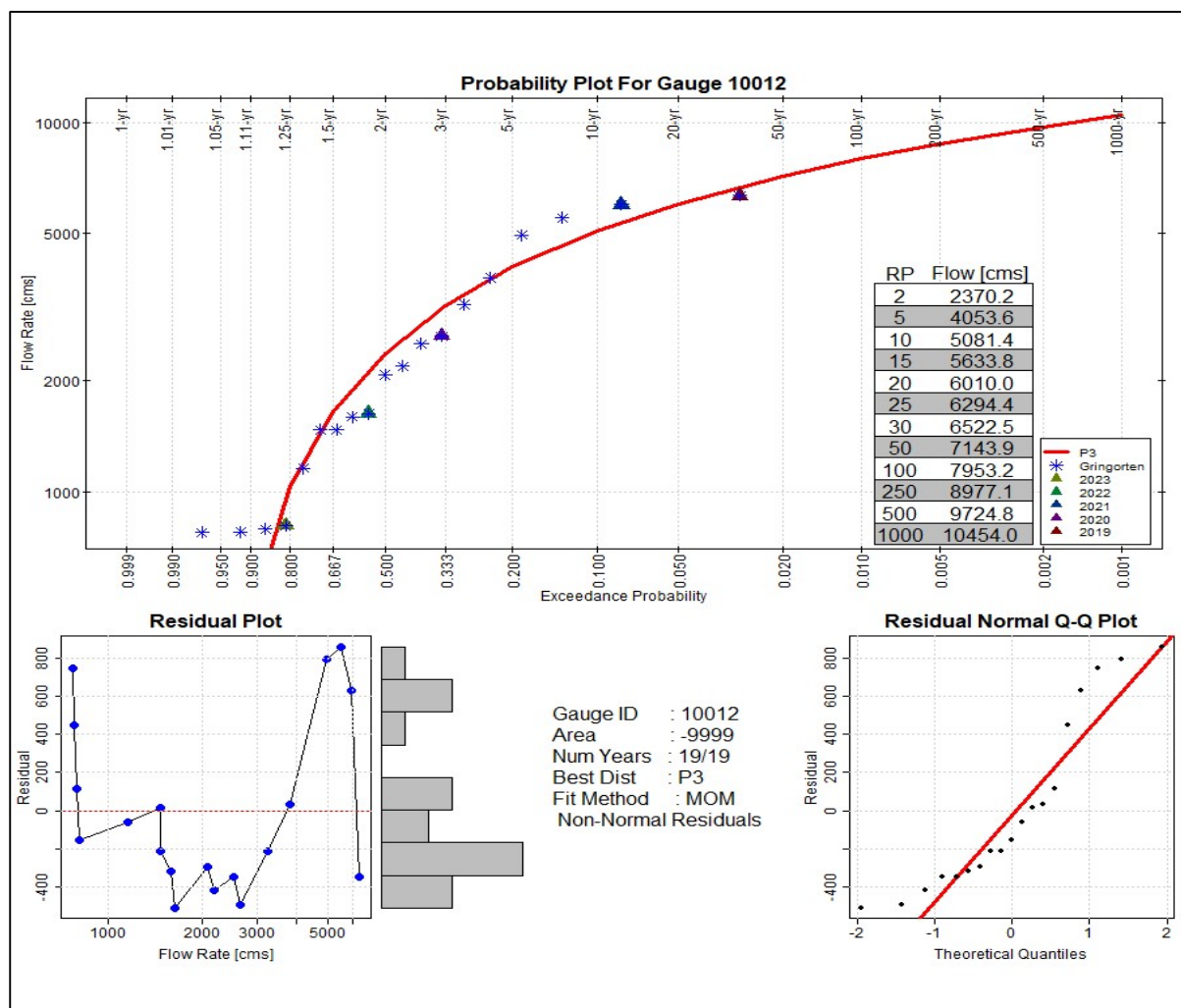


Figure 1.38 Best fit probability distribution results based on FFA approach for Sangali (Irwin Bridge).

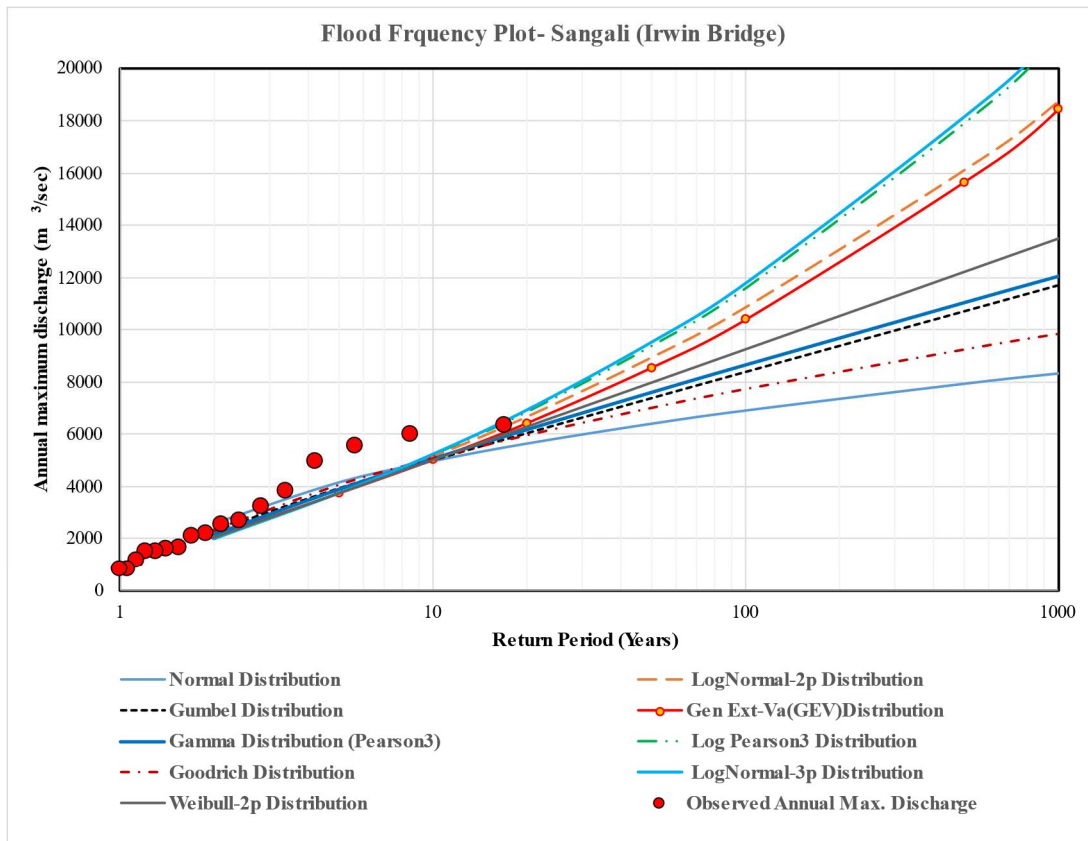


Figure 1.39 Flood frequency plot using FFD 2.1 software for Sangali (Irwin Bridge).

Table 1.17 Results of the flood frequency analysis using FFD 2.1 software at Sangali (Irwin Bridge)

Return Period (Years)	Non-Exceedence probability	Gamma (Pearson3)	Gumbel	Goodrich
		Quantile (%)	Quantile (%)	Quantile (%)
2	0.5	2195.1	2311.2	2346.7
5	0.8	3912.8	3942.5	4063.6
10	0.9	5081.1	5022.6	5080.7
20	0.95	6195.4	6058.6	5970.9
50	0.98	7617.3	7399.6	7019.7
100	0.99	8666.4	8404.6	7743.2
500	0.998	11044.6	10726.8	9259.9
1000	0.999	12050.7	11725.1	9859.3
10000	0.9999	15340.5	15039.8	11683.8

From the FFD 2.1 result (Figure 1.39), it can be observed that several probability distributions can be accepted based on various performance metrics used for selection of these probability distributions. The best fitting distributions are Gamma (Pearson type III), Gumbel, and Goodrich distributions for the Sangali (Irwin Bridge) gauging site. The results of this analysis for Sangali (Irwin Bridge) gauging site are summarised in Figure 1.39 and Table 1.17. Among these distributions, the Gamma (Pearson type III) distribution stands out as the best-fitting distribution.

The flood frequency analysis results for Sangali (Irwin Bridge) a gauging site obtained with FFD 2.1 and FFA approach are comparable. Therefore, the Pearson III distribution (MOM) obtained with FFA approach (Figure 1.38) is recommended as the best fitted probability distribution for estimating flood quantiles for different return periods at Sangali (Irwin Bridge) gauging site.

Flood Frequency Analysis: Rajaram KT Weir

According to the FFA for the Rajaram KT Weir gauging site, the three best-fitting probability distributions are the Pearson type III, Normal and Log Pearson type III distributions. The

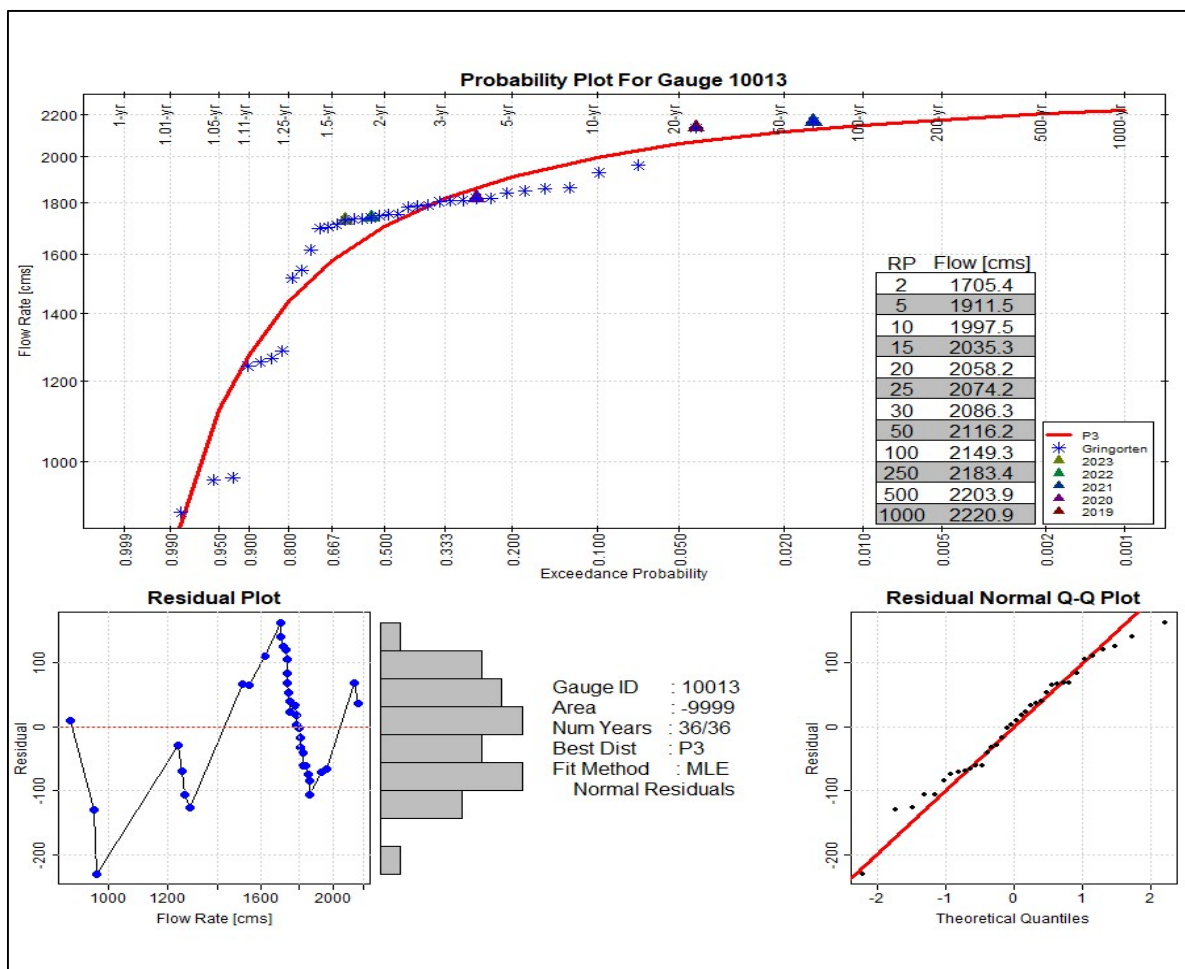


Figure 1.40 Best fit probability distribution results based on FFA approach for Rajaram KT Weir.

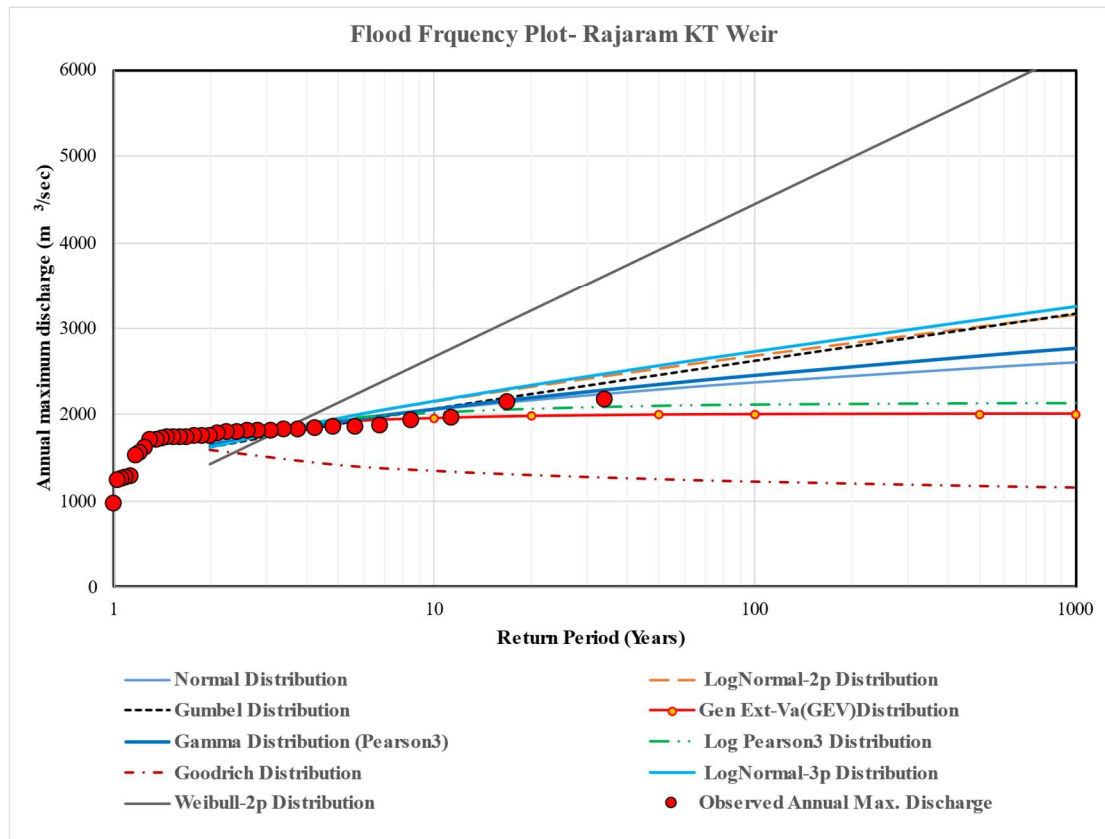


Figure 1.41 Flood frequency plot using FFD 2.1 software for Rajaram KT Weir.

Table 1.18 Results of the flood frequency analysis using FFD 2.1 software at Rajaram KT Weir

Return Period (Years)	Non-Exceedence probability	Gamma (Pearson3)	Log Pearson 3	Normal
		Quantile (%)	Quantile (%)	Quantile (%)
2	0.5	1644.5	1718.6	1663.1
5	0.8	1912.8	1939.8	1919.8
10	0.9	2063.9	2018.1	2053.9
20	0.95	2194.4	2064.7	2164.7
50	0.98	2347.6	2099.9	2289.4
100	0.99	2453.5	2114.9	2372.5
500	0.998	2677.1	2129.7	2540.8
1000	0.999	2766.4	2131.5	2605.5
10000	0.9999	3042.3	2132.4	2797.3

probability plot derived using the best fit Pearson type III distribution (MLE) for the Rajaram KT Weir gauging site is displayed in Figure 1.40. Additionally, the flood quantiles for different return periods are summarised in this chart. When comparing the specified distribution curve to the observed data, the residual plot indicates the inaccuracy in the discharge value calculation. Confidence in the choice of flood quantile for different return periods is thus given. In general, this figure offers significant information about the possible flood levels anticipated for the Rajaram KT Weir gauge location for different recurrence intervals. However, most of the probability distribution method do not provide good fit for this gauging site so results may be used with cautions.

From the FFD 2.1 result (Figure 1.41), it can be observed that several probability distributions can be accepted based on various performance metrics used for selection of these probability distributions. The best fitting distributions are Gamma (Pearson type III), Log Pearson III, and Normal distributions for the Rajaram KT Weir gauging site. The results of this analysis for Rajaram KT Weir gauging site are summarised in Figure 1.41 and Table 1.18. Among these distributions, the Gamma (Pearson type III) distribution stands out as the best-fitting distribution.

The flood frequency analysis results for Rajaram KT Weir a gauging site obtained with FFD 2.1 and FFA approach are comparable. Therefore, the Pearson III distribution (MLE) obtained with FFA approach (Figure 1.40) is recommended as the best fitted probability distribution for estimating flood quantiles for different return periods at Rajaram KT Weir gauging site. However, as mentioned earlier as the most of the probability distributions do not provide good fit for this gauging site so results may be used with cautions.

Summary of results for the considered gauging sites

The flood frequency quantiles derived for different return periods through the utilization of various probability distributions, along with the implementation of the maximum likelihood method and method of moments for estimating their parameters, are illustrated in Table 1.19 and 1.20, respectively. The results from all other distributions for all these selected gauging sites are presented in Annexure I. Through a comprehensive examination of flood frequency quantiles derived using the FFA method and a subsequent comparison with results from the FFD method, the flood frequency quantiles recommended for the selected gauging sites are detailed in Table 1.21.

Figure 1.19 Flood Frequency Quantiles for Various Return Periods (Maximum Likelihood Method)

Station name	Karad	Warunji	Samdoli (Seasonal)	Arjunwad (Seasonal)	Terwad (Seasonal)	Kurundwad	Sadalga (Seasonal)	Gokak Falls (Seasonal)	Mudhol	Koyna	Sangali (Irwin Bridge)	Rajaram KT weir
Gauge ID	10001	10002	10004	10005	10006	10007	10008	10009	10010	10011	10012	10013
Selected Distribution	LN	LN	P3	LN	GUMBEL	GUMBEL	P3	GUMBEL	GEV	LN	P3	P3
Q2	2262.40	1475.31	1436.80	3277.20	1642.57	4521.17	1121.73	1050.87	432.85	582.68	1844.61	1705.35
Q5	3689.48	2464.33	1875.61	4765.39	2352.01	5990.17	1542.85	1857.29	966.01	1595.88	3973.02	1911.54
Q10	4764.18	3222.32	2133.66	5795.49	2821.72	6962.78	1814.80	2391.20	1521.21	2702.25	5748.25	1997.51
Q15	5412.41	3683.75	2270.03	6390.03	3086.73	7511.52	1964.51	2692.44	1936.85	3514.51	6821.79	2035.26
Q20	5884.00	4021.17	2362.12	6812.05	3272.28	7895.74	2067.73	2903.35	2283.48	4174.56	7594.81	2058.20
Q25	6257.22	4289.15	2431.32	7140.43	3415.20	8191.68	2146.35	3065.81	2586.87	4738.43	8199.79	2074.23
Q30	6567.28	4512.38	2486.59	7409.75	3531.47	8432.44	2209.77	3197.98	2859.91	5234.87	8697.17	2086.32
Q50	7462.15	5159.48	2636.21	8170.98	3855.48	9103.35	2384.07	3566.27	3765.44	6810.84	10103.31	2116.20
Q75	8201.67	5697.14	2750.30	8783.95	4111.38	9633.24	2519.41	3857.16	4661.33	8274.57	11230.57	2136.45
Q100	8742.96	6092.22	2829.11	9224.41	4292.50	10008.28	2614.06	4063.04	5412.67	9439.04	12035.38	2149.27
Q150	9530.32	6669.01	2937.58	9853.87	4547.34	10535.97	2745.79	4352.71	6666.83	11274.01	13175.78	2165.52
Q200	10106.92	7092.91	3012.89	10307.11	4727.93	10909.91	2838.19	4557.99	7719.28	12724.37	13988.74	2175.91
Q250	10564.75	7430.35	3070.44	10662.67	4867.92	11199.77	2909.32	4717.11	8643.47	13940.38	14621.27	2183.40
Q500	12048.06	8528.37	3244.96	11790.86	5302.40	12099.44	3127.54	5210.99	12248.04	18273.53	16595.68	2203.90
Q750	12960.25	9206.98	3344.42	12468.38	5556.40	12625.39	3253.52	5499.71	14995.92	21238.40	17756.48	2214.23
Q1000	13628.13	9705.34	3413.95	12957.35	5736.57	12998.46	3342.24	5704.51	17303.00	23554.85	18582.37	2220.93
Q2000	15310.30	10965.75	3578.28	14164.85	6170.58	13897.16	3554.01	6197.85	24393.25	29937.52	20579.16	2235.22
Q5000	17699.59	12767.55	3789.53	15827.96	6744.20	15084.93	3830.18	6849.88	38326.73	40361.14	23231.49	2250.75
Q7500	18819.87	13616.60	3881.11	16589.33	6998.01	15610.49	3951.20	7138.39	46784.03	45800.86	24409.11	2256.61
Q10000	19639.01	14238.99	3945.45	17139.32	7178.09	15983.37	4036.67	7343.08	53885.44	50002.60	25245.94	2260.46
Q1e+05	26969.90	19860.84	4444.42	21850.08	8619.36	18967.75	4710.88	8981.36	166511.3	96116.75	31975.87	2283.72

Figure 1.20 Flood Frequency Quantiles for Various Return Periods (Method of Moments)

Station name	Karad	Warunji	Samdoli (Seasonal)	Arjunwad (Seasonal)	Terwad (Seasonal)	Kurundwad	Sadalga (Seasonal)	Gokak Falls (Seasonal)	Mudhol	Koyna	Sangali (Irwin Bridge)	Rajaram KT weir
Gauge ID	10001	10002	10004	10005	10006	10007	10008	10009	10010	10011	10012	10013
Selected Distribution	P3	LN	P3	LN	GUMBEL	GUMBEL	P3	GUMBEL	GEV	P3	P3	P3
Q2	2440.25	1475.31	1442.33	3277.20	1641.91	4527.54	1156.12	1069.98	534.97	889.93	2370.226	1609.91
Q5	3866.14	2475.40	1880.13	4781.86	2353.94	6091.89	1552.56	1956.95	1298.31	1753.92	4053.638	1891.68
Q10	4750.56	3244.39	2134.20	5826.02	2825.36	7127.63	1788.85	2544.21	1857.80	2286.53	5081.421	2072.01
Q15	5229.16	3713.31	2267.65	6429.48	3091.33	7711.98	1914.48	2875.53	2194.11	2573.98	5633.79	2170.92
Q20	5556.31	4056.54	2357.48	6858.15	3277.56	8121.13	1999.59	3107.51	2438.94	2770.21	6010.034	2238.98
Q25	5804.14	4329.31	2424.85	7191.86	3421.00	8436.29	2063.69	3286.20	2632.96	2918.73	6294.414	2290.76
Q30	6003.25	4556.65	2478.57	7465.67	3537.70	8692.67	2114.96	3431.57	2794.38	3037.97	6522.503	2332.50
Q50	6547.13	5216.20	2623.64	8240.07	3862.88	9407.12	2254.08	3836.66	3261.72	3363.37	7143.942	2447.05
Q75	6966.37	5764.77	2733.92	8864.14	4119.72	9971.41	2360.47	4156.60	3649.80	3613.91	7621.505	2535.86
Q100	7258.17	6168.14	2809.96	9312.81	4301.50	10370.79	2434.11	4383.05	3935.05	3788.14	7953.204	2597.91
Q150	7662.47	6757.46	2914.41	9954.36	4557.27	10932.73	2535.65	4701.66	4351.85	4029.39	8411.935	2684.18
Q200	7944.93	7190.85	2986.80	10416.54	4738.52	11330.94	2606.26	4927.44	4658.54	4197.81	8731.852	2744.65
Q250	8161.74	7536.01	3042.05	10779.25	4879.01	11639.62	2660.29	5102.46	4902.95	4327.04	8977.125	2791.16
Q500	8823.97	8660.06	3209.27	11930.83	5315.08	12597.68	2824.45	5645.67	5700.39	4721.44	9724.811	2933.74
Q750	9204.38	9355.41	3304.35	12622.89	5570.00	13157.77	2918.21	5963.23	6195.12	4947.81	10153.38	3015.97
Q1000	9471.53	9866.34	3370.73	13122.57	5750.83	13555.05	2983.83	6188.49	6559.47	5106.71	10453.98	3073.85
Q2000	10106.84	11159.55	3527.35	14357.25	6186.43	14512.08	3139.21	6731.11	7485.23	5484.35	11167.63	3211.88
Q5000	10930.89	13010.47	3728.13	16059.37	6762.14	15776.95	3339.44	7448.28	8820.50	5973.73	12091.07	3391.69
Q7500	11290.57	13883.48	3815.01	16839.16	7016.87	16336.62	3426.41	7765.61	9454.97	6187.19	12493.4	3470.43
Q10000	11544.11	14523.74	3875.98	17402.66	7197.61	16733.70	3487.57	7990.75	9922.32	6337.61	12776.76	3526.01
Q1e+05	13531.64	20316.97	4347.31	22235.52	8644.13	19911.78	3963.31	9792.69	14237.2	7515.48	14991.69	3963.91

Figure 1.21 The recommended flood frequency quintiles for each gauging sites in Upper Krishna basin.

Station name	Karad	Warunji	Samdoli (Seasonal)	Arjunwad (Seasonal)	Terwad (Seasonal)	Kurundwad	Sadalga (Seasonal)	Gokak Falls (Seasonal)	Mudhol	Koyna	Sangali (Irwin Bridge)	Rajaram KT weir
Gauge ID	10001	10002	10004	10005	10006	10007	10008	10009	10010	10011	10012	10013
Selected Distribution	P3	LN	P3	LN	GUMBEL	GUMBEL	P3	GUMBEL	GEV	P3	P3	P3
Parameter Estimation Method	MOM	MOM	MLE	MOM	MOM	MOM	MOM	MOM	MLE	MOM	MOM	MLE
Q2	2440.25	1475.31	1436.80	3277.20	1641.91	4527.54	1156.12	1069.98	432.85	889.93	2370.23	1705.35
Q5	3866.14	2475.40	1875.61	4781.86	2353.94	6091.89	1552.56	1956.95	966.01	1753.92	4053.64	1911.54
Q10	4750.56	3244.39	2133.66	5826.02	2825.36	7127.63	1788.85	2544.21	1521.21	2286.53	5081.42	1997.51
Q15	5229.16	3713.31	2270.03	6429.48	3091.33	7711.98	1914.48	2875.53	1936.85	2573.98	5633.79	2035.26
Q20	5556.31	4056.54	2362.12	6858.15	3277.56	8121.13	1999.59	3107.51	2283.48	2770.21	6010.03	2058.20
Q25	5804.14	4329.31	2431.32	7191.86	3421.00	8436.29	2063.69	3286.20	2586.87	2918.73	6294.41	2074.23
Q30	6003.25	4556.65	2486.59	7465.67	3537.70	8692.67	2114.96	3431.57	2859.91	3037.97	6522.50	2086.32
Q50	6547.13	5216.20	2636.21	8240.07	3862.88	9407.12	2254.08	3836.66	3765.44	3363.37	7143.94	2116.20
Q75	6966.37	5764.77	2750.30	8864.14	4119.72	9971.41	2360.47	4156.60	4661.33	3613.91	7621.50	2136.45
Q100	7258.17	6168.14	2829.11	9312.81	4301.50	10370.79	2434.11	4383.05	5412.67	3788.14	7953.20	2149.27
Q150	7662.47	6757.46	2937.58	9954.36	4557.27	10932.73	2535.65	4701.66	6666.83	4029.39	8411.93	2165.52
Q200	7944.93	7190.85	3012.89	10416.54	4738.52	11330.94	2606.26	4927.44	7719.28	4197.81	8731.85	2175.91
Q250	8161.74	7536.01	3070.44	10779.25	4879.01	11639.62	2660.29	5102.46	8643.47	4327.04	8977.12	2183.40
Q500	8823.97	8660.06	3244.96	11930.83	5315.08	12597.68	2824.45	5645.67	12248.04	4721.44	9724.81	2203.90
Q750	9204.38	9355.41	3344.42	12622.89	5570.00	13157.77	2918.21	5963.23	14995.92	4947.81	10153.38	2214.23
Q1000	9471.53	9866.34	3413.95	13122.57	5750.83	13555.05	2983.83	6188.49	17303.00	5106.71	10453.98	2220.93
Q2000	10106.84	11159.55	3578.28	14357.25	6186.43	14512.08	3139.21	6731.11	24393.25	5484.35	11167.63	2235.22
Q5000	10930.89	13010.47	3789.53	16059.37	6762.14	15776.95	3339.44	7448.28	38326.73	5973.73	12091.07	2250.75
Q7500	11290.57	13883.48	3881.11	16839.16	7016.87	16336.62	3426.41	7765.61	46784.03	6187.19	12493.40	2256.61
Q10000	11544.11	14523.74	3945.45	17402.66	7197.61	16733.70	3487.57	7990.75	53885.44	6337.61	12776.76	2260.46
Q1e+05	13531.64	20316.97	4444.42	22235.52	8644.13	19911.78	3963.31	9792.69	166511.3	7515.48	14991.69	2283.72

1.5 Conclusion

Flood frequency analysis is a critical component of hydrological studies, providing essential data for the development of effective flood risk management and mitigation strategies. The first step in flood frequency analysis employed was the pre-processing of the available annual maximum discharge timeseries data. Further, the trend analysis has been carried and shift and trend in the data were analysed. Further the statistical properties of the timeseries data at considered gauging sites has been carried out. Based on this analysis the brief conclusions are as follows:

- Mean of sample at gauging site Kurundwad is higher as compared to other gauging sites which suggests that flood magnitudes is larger. The maximum mean value of 4818.3 Cumec observed at Kurundwad whereas minimum mean value of 731.2 Cumec observed at Mudhol.
- Variance of sample estimated at gauging sites Sangali, Kurundwad, Arjunwad, Karad and Warunji, respectively are higher as compared to other gauging sites which suggests that flood magnitudes vary widely at these gauging stations. The variance estimated for Terwad station is higher than Samdoli. A variance of sample estimated at Rajaram KT weir and Sadalga gauging sites are low which means that the flood magnitudes are relatively consistent at these gauging stations.
- Standard deviation of sample estimated at gauging sites Sangali, Kurundwad, Arjunwad, Karad and Warunji, respectively are higher as compared to other gauging sites which suggests flood magnitudes vary widely at these gauging stations. A standard deviation of sample estimated at Rajaram KT weir and Sadalga gauging sites are low which means that the flood magnitudes are relatively consistent at these gauging stations.
- The maximum discharge values observed at Kurundawad, Arjunwad, Karad and Sangali (Irwin bridge) gauging sites were 10092, 9381, 6708.2 and 6327.8 Cumec, respectively. Whereas the maximum discharge values observed at Terwad, Samdoli and Sadalga were 4110, 3064 and 2435 Cumec, respectively.
- The minimum discharge values observed at Karad, Sangali (Irwin bridge), Arjunwad and Kurundawad gauging sites were 677.6, 775.5, 1332.6, and 2275 Cumec, respectively. Whereas the minimum discharge values observed at Samdoli, Terwad, and Sadalga were 4.1, 444.3, and 537.7 Cumec, respectively.
- The Rajaram KT Weir and Samdoli gauging sites have negative skewness which indicates a longer left tail, with more extreme low values. The Mudhol, Warunji, Kurundwad, Arjunwad, Terwad and Karad gauging sites, respectively shows positive skewness with higher values that mean they have a longer right tail, with more extreme high flood values.

- The Mudhol, Warunji and Kurundwad, Arjunwad and Terwad gauging sites has high kurtosis value indicates more frequent extreme values as compared to other gauging sites.

Further, the present study conducted in the Upper Krishna Basin (UKB) exhibit the application of various statistical methods to analyse annual peak flow data at various gauging sites. The comprehensive analysis of flood frequency (FFA) for the selected gauging locations in UKB was conducted utilizing different probability distributions (Pearson III (P3), Log Normal (LN), Log Pearson III (LP3), Generalized Extreme Value (GEV), Gumbel and Normal distributions) and various plotting position techniques (Weibull, Blom, Cunnane, Gringorten, and Hazen) within the R programming environment. The selection of distinct probability distributions at each gauging site was based on the application of Goodness-of-Fit (GoF) criteria such as the Akaike Information Criterion (AIC), Corrected Akaike Information Criterion (AICc), Bayesian Information Criterion (BIC), Anderson-Darling Criterion (ADC), Modified Anderson-Darling (AD) Criterion, Kolmogorov-Smirnov (KS), and Comb Method. Alternatively, the estimation of flood frequency analysis outcomes was performed through the utilization of Flood Frequency Distribution (FFD 2.1) (such as Normal, Log Normal (2p), Gumbel, General Extreme Values Distribution (GEV), Gamma (Pearson III), Log Pearson 3, Goodrich, Log Normal (3p) and Weibull distributions) in MATLAB software. Subsequently, the results obtained from FFD 2.1 were employed to validate the findings derived from the FFA conducted in R programming.

The selection process for the best-fit statistical distributions is both methodical and empirical, involving visual inspections and a series of goodness-of-fit tests. This rigorous approach ensures that the chosen models accurately reflect the flood frequency characteristics of the UKB. The best fitted probability distribution for each gauging sites is described in the previous section for each gauging site. The best flood frequency quantiles for various return periods based on best fitting probability distributions for various gauging sites using parameter estimation methods such as maximum likelihood method (MLE) and method of moment (MOM) are presented in Tables 1.19 and 1.20, respectively. The flood frequency quantiles recommended for the gauging sites within the Upper Krishna Basin, as determined through the FFA approach, have been outlined in Table 1.21. These quantiles could potentially be employed in the implementation of flood mitigation efforts.

Annexure-II

II. The Annual Maximum Discharge Series data used for each gauging station in this study for flood frequency analysis.

Station ID	10001	10002	10003	10004	10005	10006	10007	10008	10009	10010	10011	10012	10013
Station Name	Karad	Warunji	Nandre	Samdoli (Seasonal)	Arjunwad (Seasonal)	Terwad (Seasonal)	Kurundwad	Sadalga (Seasonal)	Gokak Falls (Seasonal)	Mudhol	Koyna	Sangali (Irwin Bridge)	Rajaram KT Weir
Lat	17.29	17.27	16.95	16.86	16.78	16.68	16.68	16.56	16.19	16.32	17.38	16.74	16.86
Long	74.19	74.17	74.53	74.50	74.63	74.58	74.60	74.53	74.79	75.20	73.73	74.24	74.56
Area	5462	1890	3035	1948	12660	2425	15190	2322	2770	6734	920		
Datum (Zero Gauge, m)	549.915	549.437	526.5	528.594	523.225	520	519.455	525.16	536	530	568		
Available Years	59	58	6	57	55	45	52	55	53	17	36	19	36
Y1965	4760.00												
Y1966	2673.30	530.00											
Y1967	4482.40	3830.40		2000.00									
Y1968	1526.50	757.40		1068.10									
Y1969	4266.80	1604.50		1650.00	3850.00			1225.00					
Y1970	2427.60	1492.30		1554.90	5078.70			1320.00					
Y1971	2155.90	1385.60		1220.40	2935.60			1189.60	929.10				
Y1972	2190.00	1522.80		1219.30	3988.80		4239.00	1230.00	2320.50		66.50		
Y1973	3050.60	1899.60		2250.30	4950.00		5659.10	1396.90	1921.80		512.10		
Y1974	3101.40	1582.50		1528.00	2468.80		3436.20	1304.10	2150.00		85.50		
Y1975	3231.40	1680.70		2049.80	5270.00		5781.20	1759.00	3106.30		1070.30		
Y1976	4561.80	2780.30		2006.90	4890.00		6853.50	1103.00	1701.80		1925.00		
Y1977	3421.80	1610.00		1548.20	3460.40		4953.10	1128.60	1194.80		1661.60		
Y1978	2187.10	1575.70		1417.90	2578.70		3583.90	1244.30	1995.20		1429.70		

Y1979	2434.50	1316.50		1435.50	4903.70	870.00	6070.20	1428.70	3147.00		811.90		
Y1980	2848.30	1868.30		1747.40	3800.00	1917.60	5174.50	1102.90	1725.00		1833.70		
Y1981	1941.60	1262.30		1740.60	3362.30	1790.00	5166.10	1457.20	1108.60		454.70		
Y1982	1261.10	1156.00		1471.80	2241.30	1430.60	3437.80	1368.20	1668.80		252.10		
Y1983	1930.50	1492.70		1681.70	3345.30	1545.00	5111.40	2435.00	2766.00		1050.90		
Y1984	1796.70	1333.80		1715.10	3284.10	1795.00	4427.00	1514.30	708.90		747.20		
Y1985	1363.30	1053.80		1530.30	3084.70	1570.00	4413.80	1421.30	716.20		147.80		
Y1986	1801.40	1283.70		1346.90	2867.60	2050.00	4587.00	1048.40	439.80		941.40		
Y1987	1323.00	1073.00		1080.00	2351.00	1392.00	3290.00	872.60	269.80		196.10		
Y1988	3205.00	2590.00		2025.00	4997.00	2080.00	4870.00	1810.00	2064.00		1000.00		1790.00
Y1989	2904.00	1600.00		2412.00	4954.00	2205.00	6000.00	2100.00	1455.00		269.50		1961.00
Y1990	4361.00	2530.00		1675.00	6500.00	2250.00	5760.00	1452.00	1073.00		2371.00		1782.00
Y1991	3350.00	1800.00		2081.00	5938.00	2553.00	6322.00	1379.00	1351.00		1317.00		1820.00
Y1992	2127.00	1238.00		1178.00	2928.00	1306.00	3919.00	773.10	871.40		428.30		1264.00
Y1993	1812.00	1023.00		1463.00	2843.00	1447.00	4051.00	1058.00	1444.00		565.90		1754.00
Y1994	3915.00	2675.00		2235.00	6300.00	2680.00	5730.00	1900.00	3440.00		2505.00		1862.00
Y1995	1356.00	772.10		869.00	2550.00	1170.00	2796.00	710.90	395.20		117.30		1286.00
Y1996	2944.00	1566.00		1610.00	3560.00	1900.00	5000.00	1180.00	728.40		158.00		1516.00
Y1997	5954.00	3529.00		1710.00	4780.00	3590.00	6800.00	1350.00	3375.00		2592.00		1805.00
Y1998	1541.00	828.00		897.50	2411.00	1051.00	3000.00	685.50	404.00		551.00		1242.00
Y1999	2111.00	1397.00		1199.00	3193.00	1540.00	3725.00	1110.00	1323.00		1453.00		1808.00
Y2000	774.10	676.70		1001.00	1747.29	890.00	2852.64	853.60	483.40	327.40	180.00		1618.00
Y2001	911.21	623.30		637.20	1763.82	1150.00	2593.78	594.60	332.70	414.10	275.00		1253.00
Y2002	1120.64	830.28		694.60	1514.00	1443.00	3014.38	827.90	475.30	418.80	752.30		1713.00
Y2003	936.54	626.70		772.10	1332.64	725.26	2275.00	537.70	210.40	213.90	65.32		890.00
Y2004	4162.90	2715.62		1340.80	4211.00	1832.00	4650.30	1287.00	558.70	643.40	2793.26		1699.00
Y2005	6312.00	4641.00		3064.00	9381.00	3340.00	10092.00	2200.00	3245.28	2100.00	3174.00	5513.28	1928.00
Y2006	6708.25	4972.54		2010.00	7505.17	2796.56	8819.46	1978.08	1702.69	1417.57	3020.59	4955.44	1842.00
Y2007	3867.81	2243.09		1568.65	3943.23	1820.56	5673.26	970.00	1990.81		53.88	3211.21	1786.00
Y2008	2884.03	1887.24		1403.12	3357.05	1951.57	5743.28	1170.00	1013.00			2186.45	1860.00

Y2009	1258.34	712.47		1141.26	2223.31	444.28	3782.14	762.60	677.40			1158.04	1704.00
Y2010	1188.74	769.77		1145.12	2587.45	1307.99	3846.78	850.90	498.50			1471.65	1754.00
Y2011	5073.85	3124.73		1591.63	4097.13	1241.68	4533.81	832.50	1123.00			3793.52	1735.00
Y2012	993.41	723.63		688.93	1614.36	962.08	2681.10	559.50	328.60			792.76	964.00
Y2013	2506.90	1924.56		1483.10	3740.79	1854.07	5465.55	847.30	621.28			2510.88	1735.00
Y2014	1699.62	1047.27		1114.85	2605.88	968.27	4197.11	710.50	451.61			1593.27	1545.00
Y2015	1046.12	235.65		669.23	1459.50	602.31	2521.54	555.40	342.60	533.47		775.54	958.00
Y2016	1261.81	1533.07		1351.00	4399.60	2200.00	4982.60	1663.06	540.28	282.00		2072.79	1850.00
Y2017	1317.86	1200.47		965.15	1831.00	1557.00	3250.00	720.00	470.00	461.56		778.32	1747.00
Y2018	1367.88	1793.84	23.26	1388.01	3095.60	2100.00	4268.19	1129.26	700.00	510.37		1471.65	1810.00
Y2019	6138.74	6003.71	123.00	2240.00	5500.00	4110.00	10040.00	1800.00	3700.00	3585.30		6327.84	2136.00
Y2020	2200.33	1989.71	680.00	1490.00	2475.80	1975.00	4303.90	980.40	0.00	382.65		2640.65	1820.00
Y2021	5611.03	5275.91	44.00	2080.00	5335.00	3620.00	8750.00	1550.00	42.59	60.80		5983.34	2165.00
Y2022	1077.88	1177.97	338.22	1282.21	1751.50	1855.00	4608.32	1000.00	61.45	322.71		1627.06	1740.00
Y2023	677.63	819.39	34.98	909.34	1555.02	963.71	3452.17	821.30	84.92	24.74		810.00	1730.00